# Personal Finance Assistant — User Acceptance Testing (UAT)

Oct 25, 2026 · @weam

## 1. Scope and environment

This UAT checks that the Personal Finance Assistant does what the research proposal and the design documents promise, used through its real interface. Each test case was run in a browser against the running app, and a screenshot was taken as evidence. Defects found were fixed and re-tested.

| Item | Value |
|---|---|
| Build | Branch `feature/demo-polish`, 251 automated tests passing |
| Interface | Streamlit 1.66, Chromium, 1400 px desktop and 390 px phone width |
| API | FastAPI on port 8000, SQLite |
| Data | **Load demo data**: sample user plus profiles A–E and a new user; amounts in SDG |
| AI assistant | Ollama 0.40.11 with Qwen 2.5 3B, temperature 0 (repeatable answers) |
| Languages | English and Arabic (right-to-left) |
| Tester | Claude, for the project team; to be repeated by the team and signed off by the supervisor |

**Entry criteria:** the app starts, demo data loads, and the automated tests pass.

**Exit criteria:** every test case passes or is formally blocked with a reason, every defect found is fixed and re-tested, and no open defect stops a demo.

## 2. Test summary

27 of 28 test cases pass and 1 fails. The failure is UAT-28: one Arabic chat question, a limit of the small 3B model, which now gets a clear "please rephrase" message. Sixteen defects were found and fixed during UAT (section 4). Nine of them were found only once the real AI model was installed and tested.

| ID | Area | Scenario | Requirement | Status |
|---|---|---|---|---|
| UAT-01 | Setup | Load demo data from the sidebar | Synthetic data | Pass |
| UAT-02 | Dashboard | The month in one sentence and the income split | Understand spending | Pass |
| UAT-03 | Spending | An expense with amount 0 is rejected | Validation | Pass |
| UAT-04 | Spending | Add an expense; the form clears | Track spending | Pass |
| UAT-05 | Spending | Edit an expense | Track spending | Pass |

| ID | Area | Scenario | Requirement | Status |
|---|---|---|---|---|
| UAT-06 | Spending | Delete an expense | Track spending | Pass |
| UAT-07 | Spending | Import a CSV with one bad row | Import a list of transactions (proposal scope) | Pass |
| UAT-08 | Income | Add income | Track income | Pass |
| UAT-09 | Bills | Add a recurring bill | Budgeting | Pass |
| UAT-10 | Analysis | Categories, trends, income and forecast tabs | Understand spending | Pass |
| UAT-11 | Budget | Budget vs spent this month | Budgeting | Pass |
| UAT-12 | Goals | Create a goal, add savings, delete it | Goals | Pass |
| UAT-13 | Goals | "Can I afford it?" for the sample user | Supervisor: advice about the future | Pass |
| UAT-14 | Goals | Same question for profile C, then a what-if | Personalized advice | Pass |
| UAT-15 | Goals | Drag a goal to the top; the plan follows it | Kanban ordering | Pass |
| UAT-16 | Essentials | Drag Restaurant to Essential; the plan stops cutting it | Kanban ordering; essential spending protected | Pass |
| UAT-17 | Plan | Steps with amounts, protected items, model confidence | Answer with a reason (RQ4) | Pass |
| UAT-18 | Money tips | Search returns a sourced note | RAG with a cited source (RQ2) | Pass |
| UAT-19 | Profiles | Create a profile and enter the basics | New users (onboarding) | Pass |
| UAT-20 | Profiles | A new user's answer says "based on what you entered" | Honest cold start | Pass |
| UAT-21 | Profiles | The chosen profile survives a language switch | Usability | Pass |
| UAT-22 | Language | The whole interface in Arabic, right-to-left | Arabic support | Pass |
| UAT-23 | Mobile | Every page at phone width, no sideways scrolling | Responsive interface | Pass |
| UAT-24 | Assistant | Clear message when the AI is offline | Reliability | Pass |
| UAT-25 | Assistant | Answers list their tools and sources (English) | Reason and source shown (RQ4) | Pass |
| UAT-26 | Assistant | A live-data question is refused | No hallucination (RQ2) | Pass |
| UAT-27 | All | Every amount in Sudanese pounds (SDG, ج.س) | Target users in Sudan | Pass |
| UAT-28 | Assistant | Four questions in Arabic in one conversation | Arabic support | Fail |

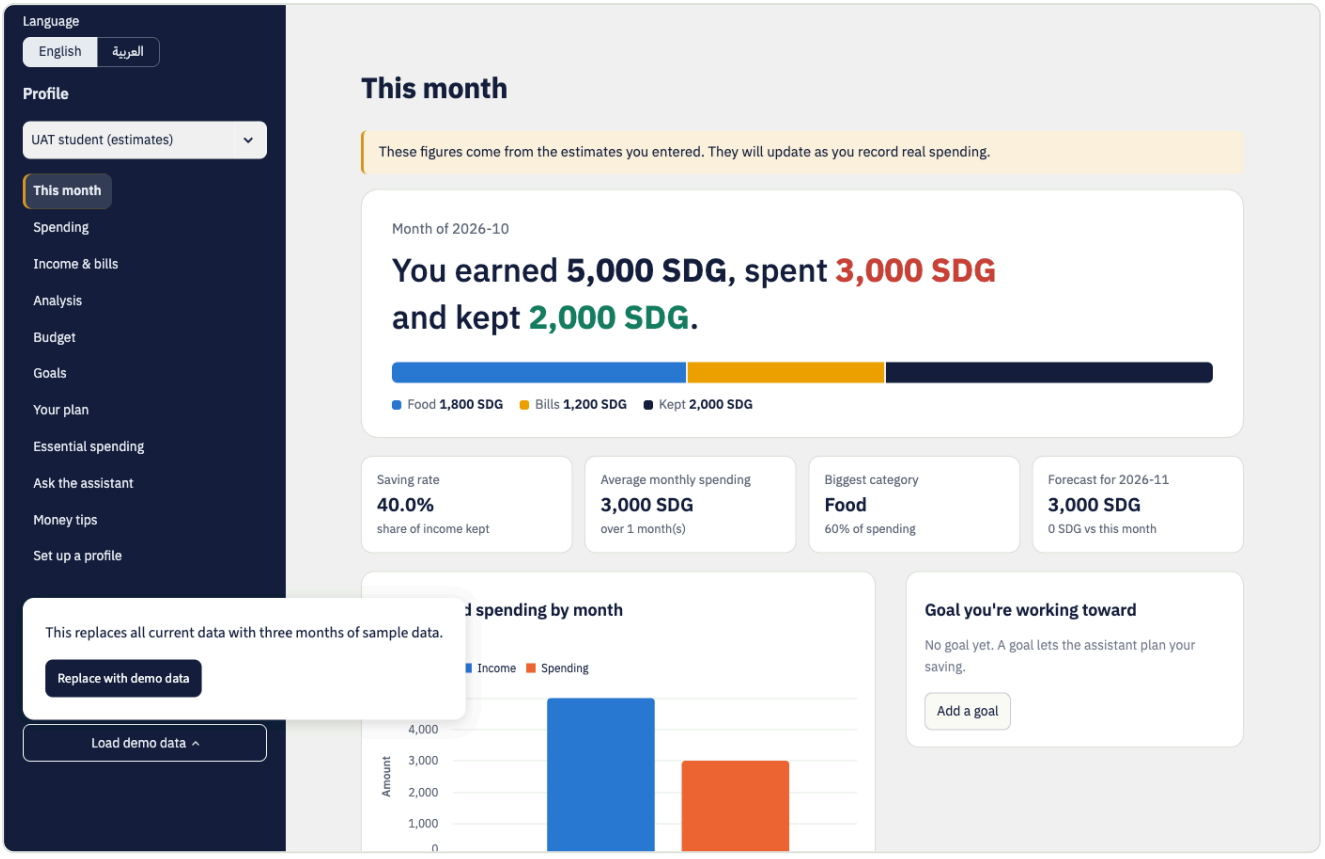
## 3. Test cases and evidence

Each case lists its steps, the expected result, the actual result and a screenshot taken during the test. Where the app stores data, the result was also checked in the database through the API.

### UAT-01 Load demo data

1. In the sidebar, open **Load demo data**.
2. Read the warning and click **Replace with demo data**.

**Expected:** all data is replaced by the sample user and the demo profiles; the dashboard reopens. **Actual:** 7 profiles exist, the test profile created earlier is gone, and the dashboard shows the sample user. Pass.

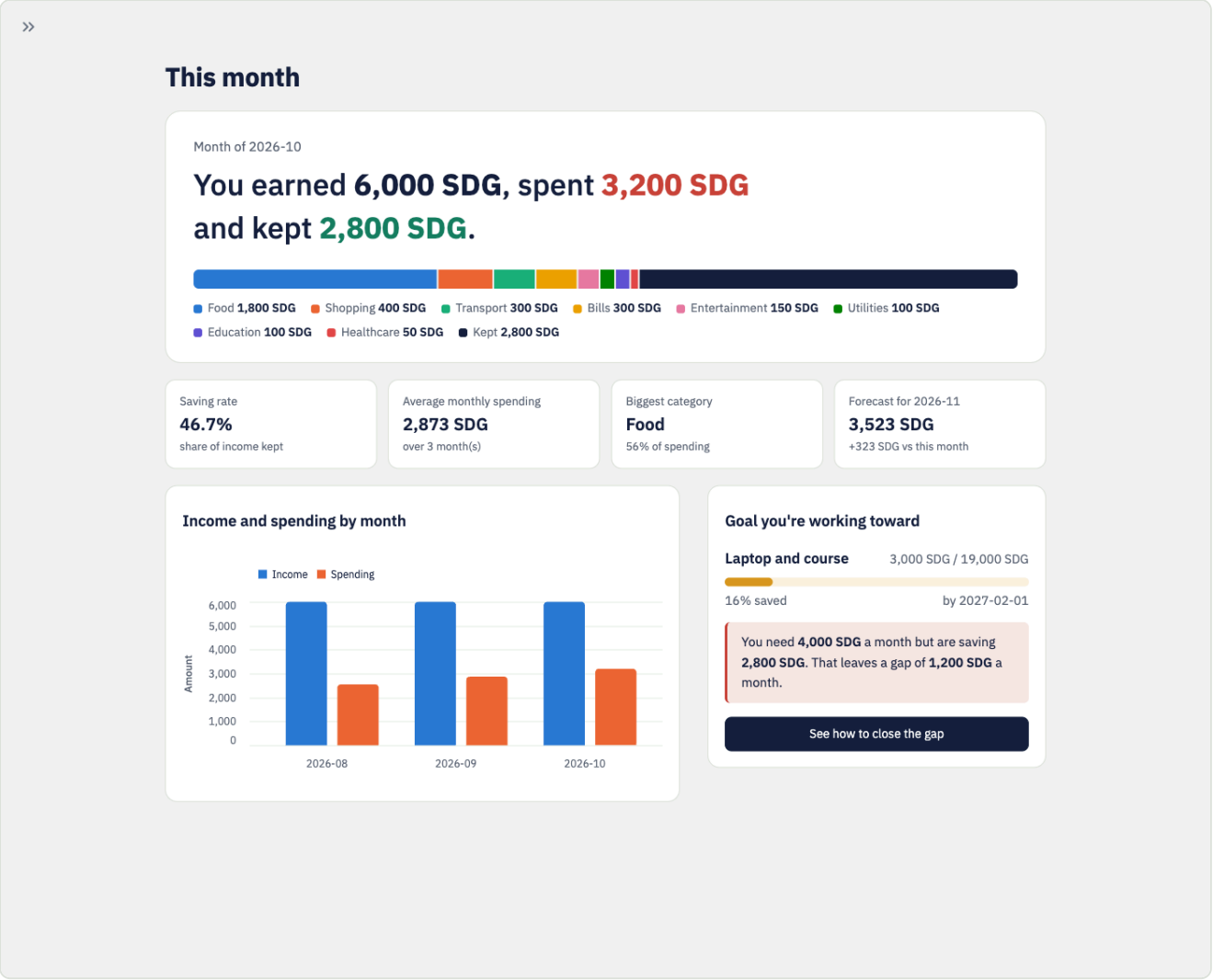



Load demo data confirmation

### UAT-02 Dashboard

1. Open **This month** for the sample user.

**Expected:** one sentence with income, spending and money kept; a bar splitting the income; four facts; a monthly chart; the goal panel. **Actual:** "You earned 6,000 SDG, spent 3,200 SDG and kept 2,800 SDG." The saving rate is 46.7% and the forecast 3,523 SDG. The goal panel shows a 1,200 SDG monthly gap. Pass.

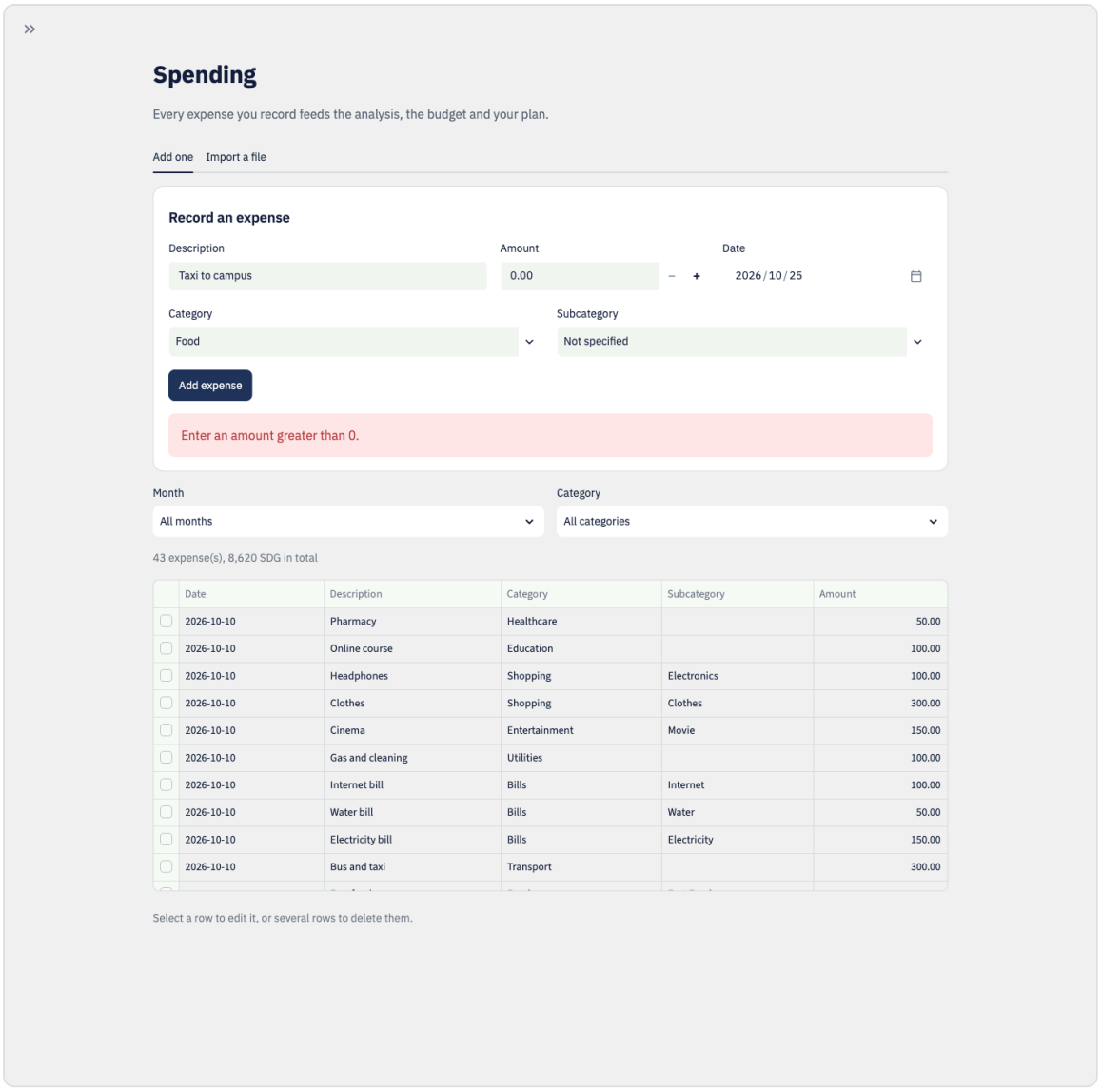



Dashboard

## UAT-03 Expense validation

1. Open **Spending → Add one** and type "Taxi to campus" with amount 0.
2. Click **Add expense**.

**Expected:** the expense is refused with a clear message. **Actual:** "Enter an amount greater than 0." is shown and nothing is saved. Pass.

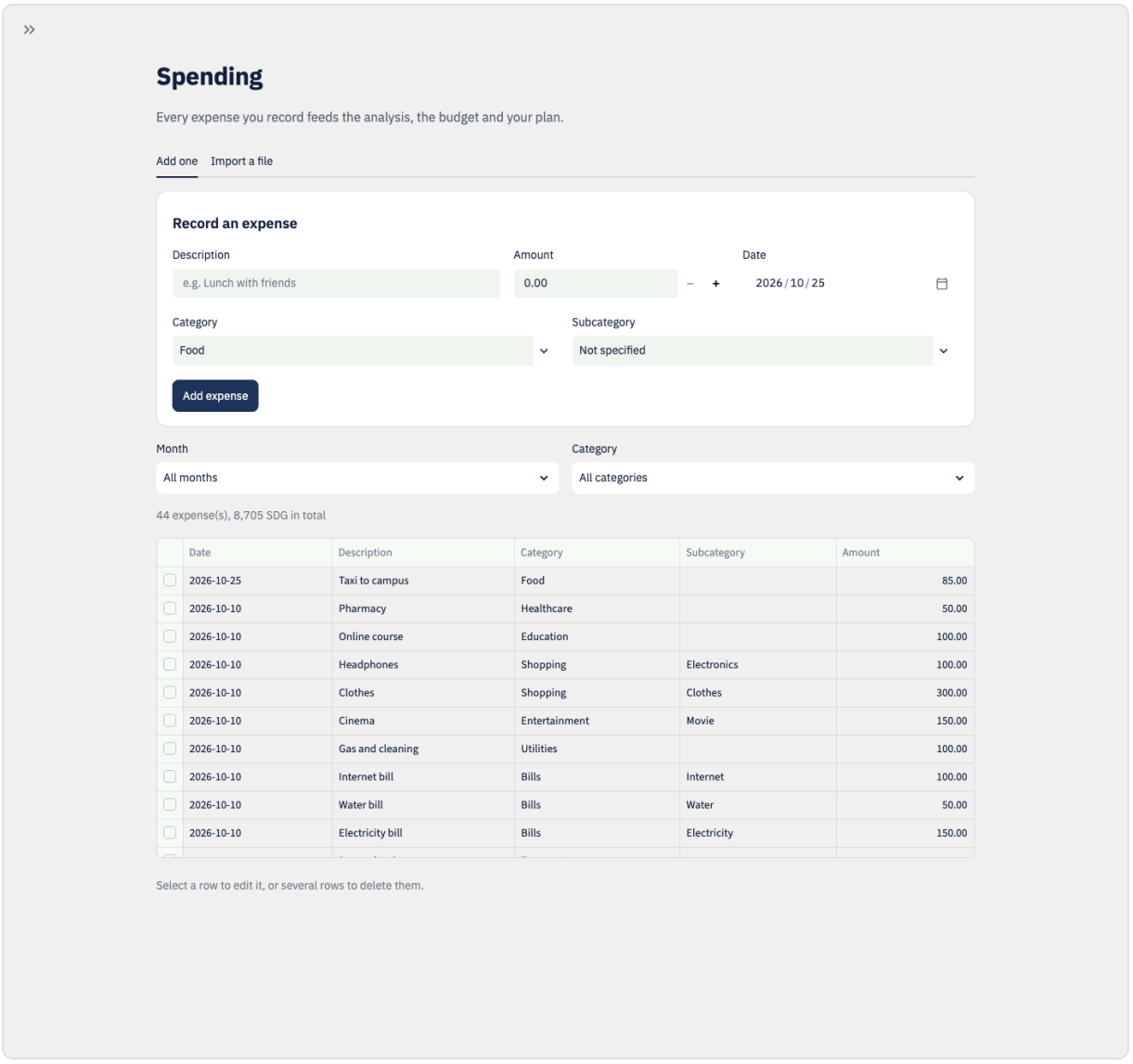



Validation message

## UAT-04 Add an expense

1. Enter amount 85, category Food, and click **Add expense**.

**Expected:** the expense is saved, appears at the top of the list, and the form is empty again. **Actual:** saved (checked through the API) and shown first in the list, dated today. The form cleared after defect D-02 was fixed. Pass.

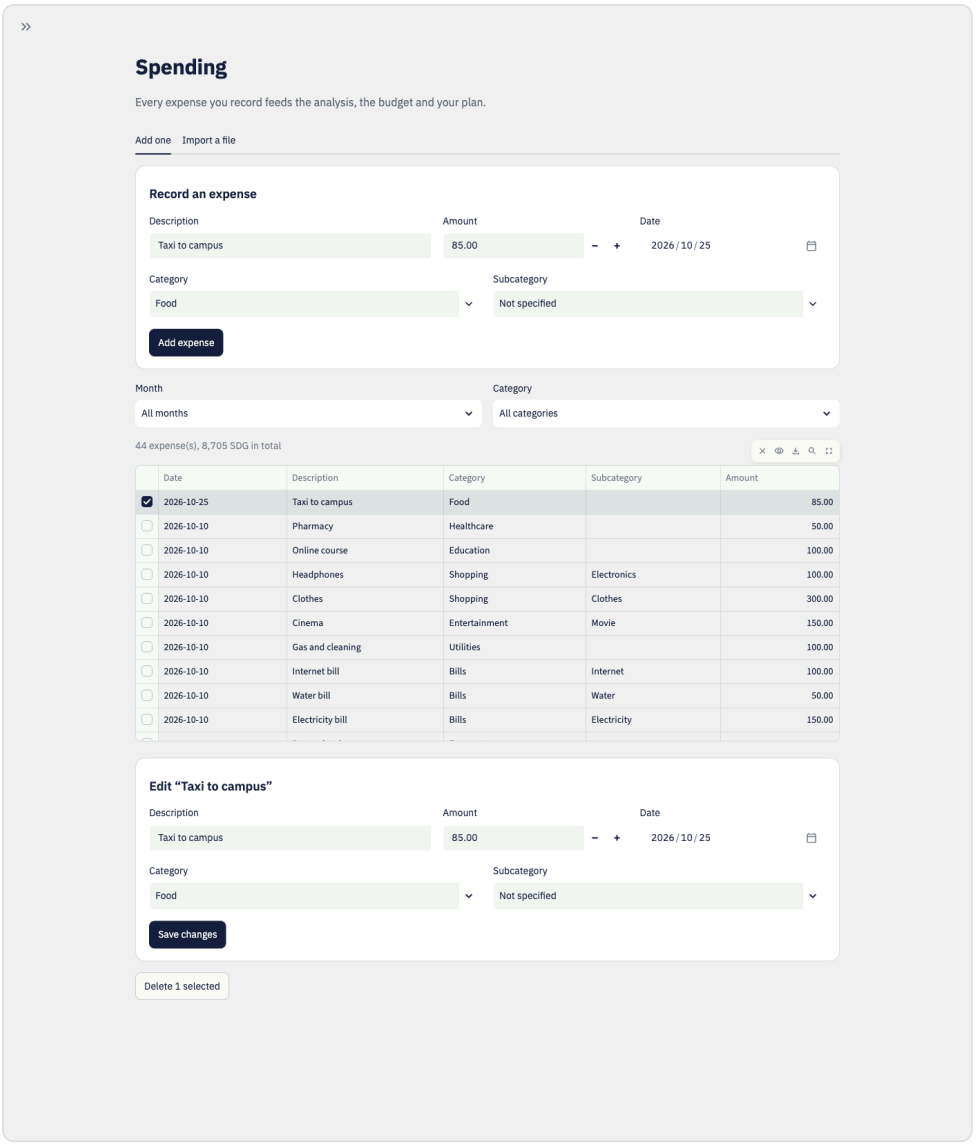



Expense added and form cleared

## UAT-05 Edit an expense

1. Tick the "Taxi to campus" row.

**Expected:** an edit form opens with the expense's values. **Actual:** "Edit “Taxi to campus”" opens with 85.00, Food and today's date, plus **Save changes** and **Delete 1 selected**. Pass.

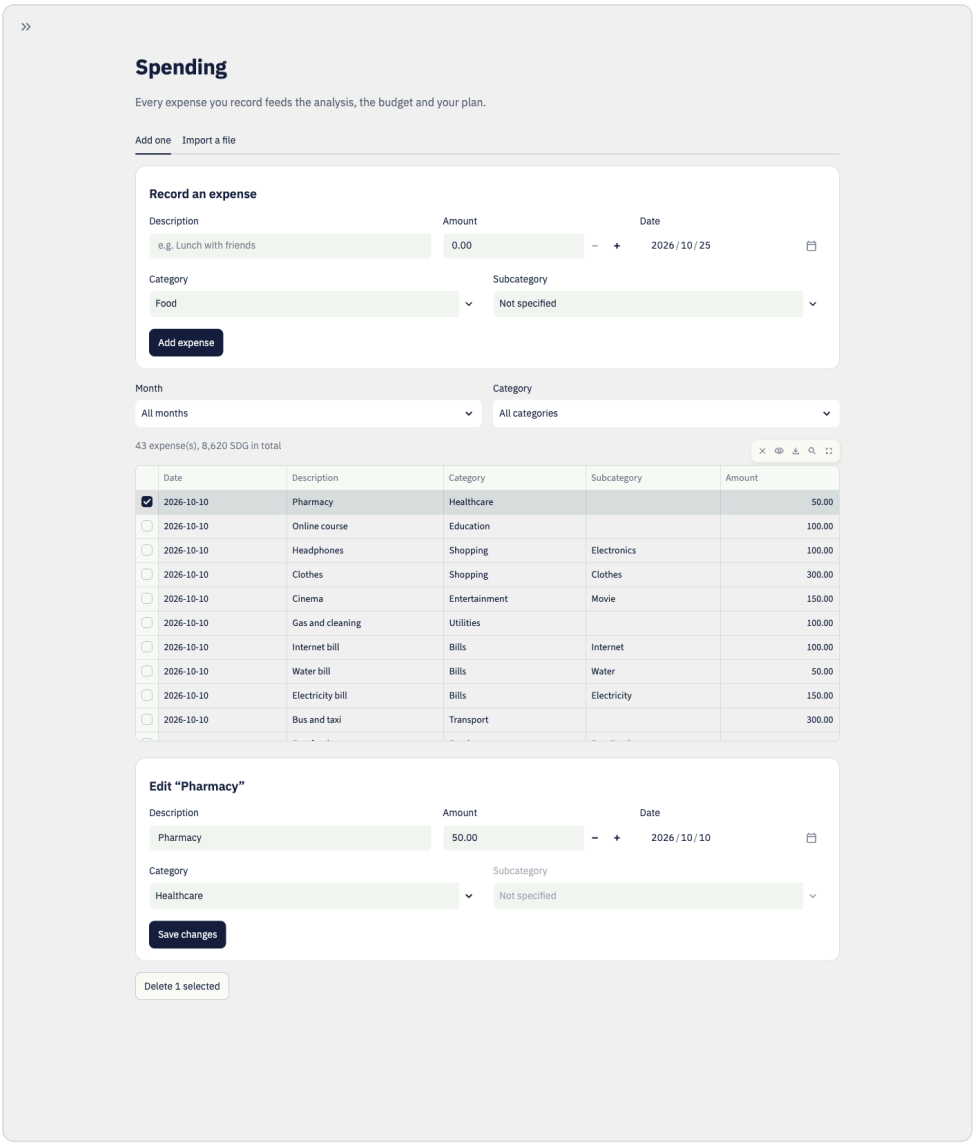



Edit form

## UAT-06 Delete an expense

1. With the row ticked, click **Delete 1 selected**.

**Expected:** the expense is removed. **Actual:** it is gone from the list and from the database. Pass.

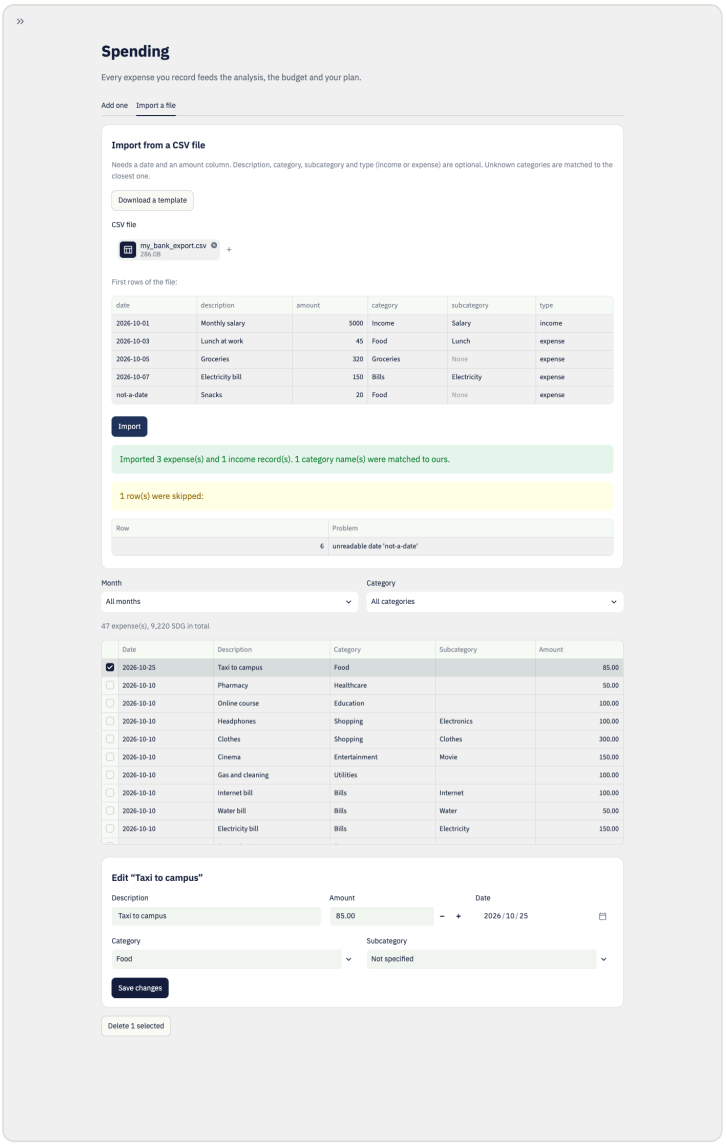



After deleting

## UAT-07 Import a CSV file

1. Open **Spending → Import a file**.
2. Upload `my_bank_export.csv`: the template plus one row with the date "not-a-date".
3. Click **Import**.

**Expected:** valid rows are imported, unknown categories are mapped, and the bad row is reported.
**Actual:** "Imported 3 expense(s) and 1 income record(s). 1 category name(s) were matched to ours" (Groceries became Food). Row 6 is reported as "unreadable date 'not-a-date'". Pass.

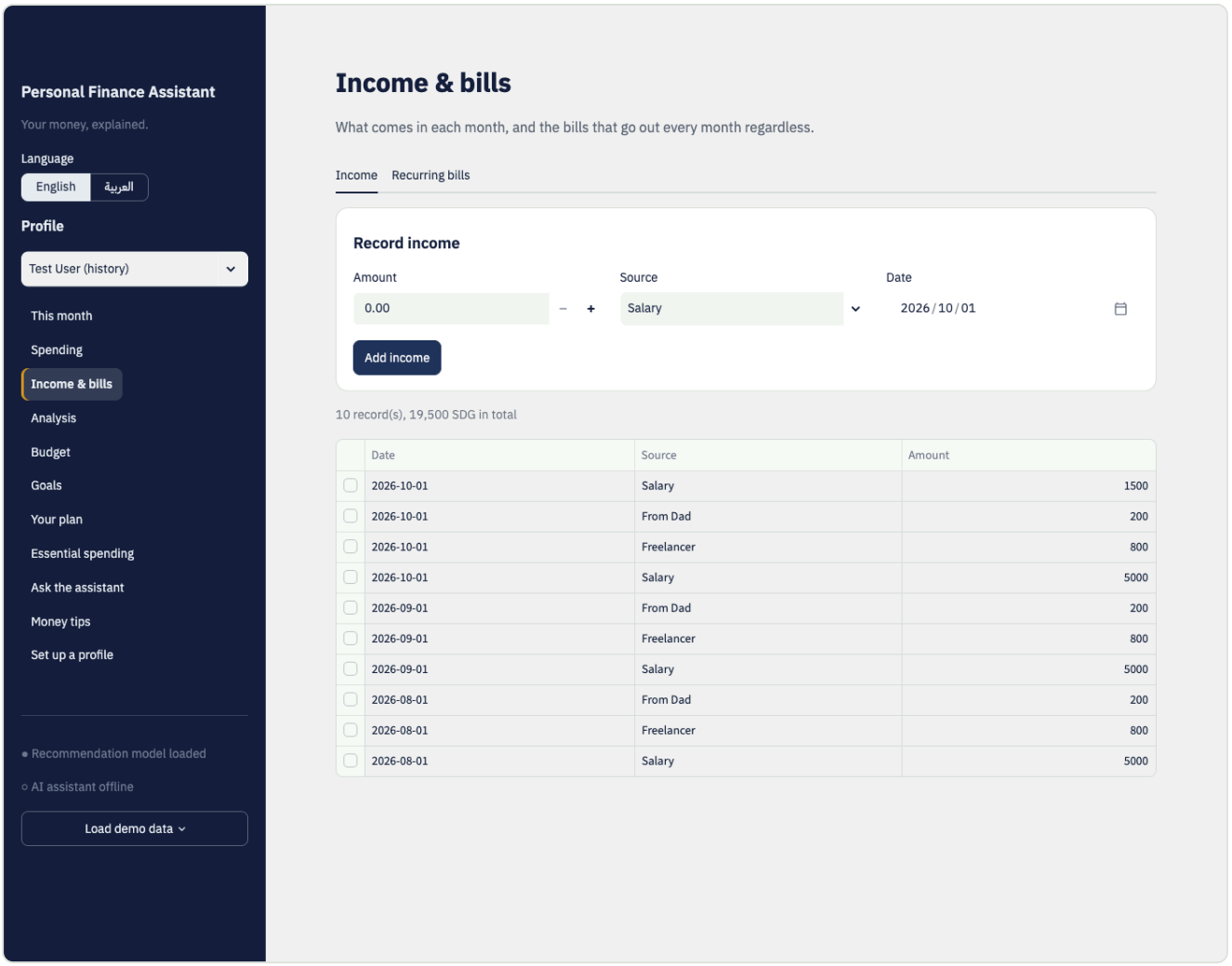



CSV import result

## UAT-08 Add income

1. Open **Income & bills → Income**, enter 1,500 and click **Add income**.

**Expected:** saved, listed, and the form cleared. **Actual:** saved (checked through the API), listed, and the amount field back at 0.00. Pass.

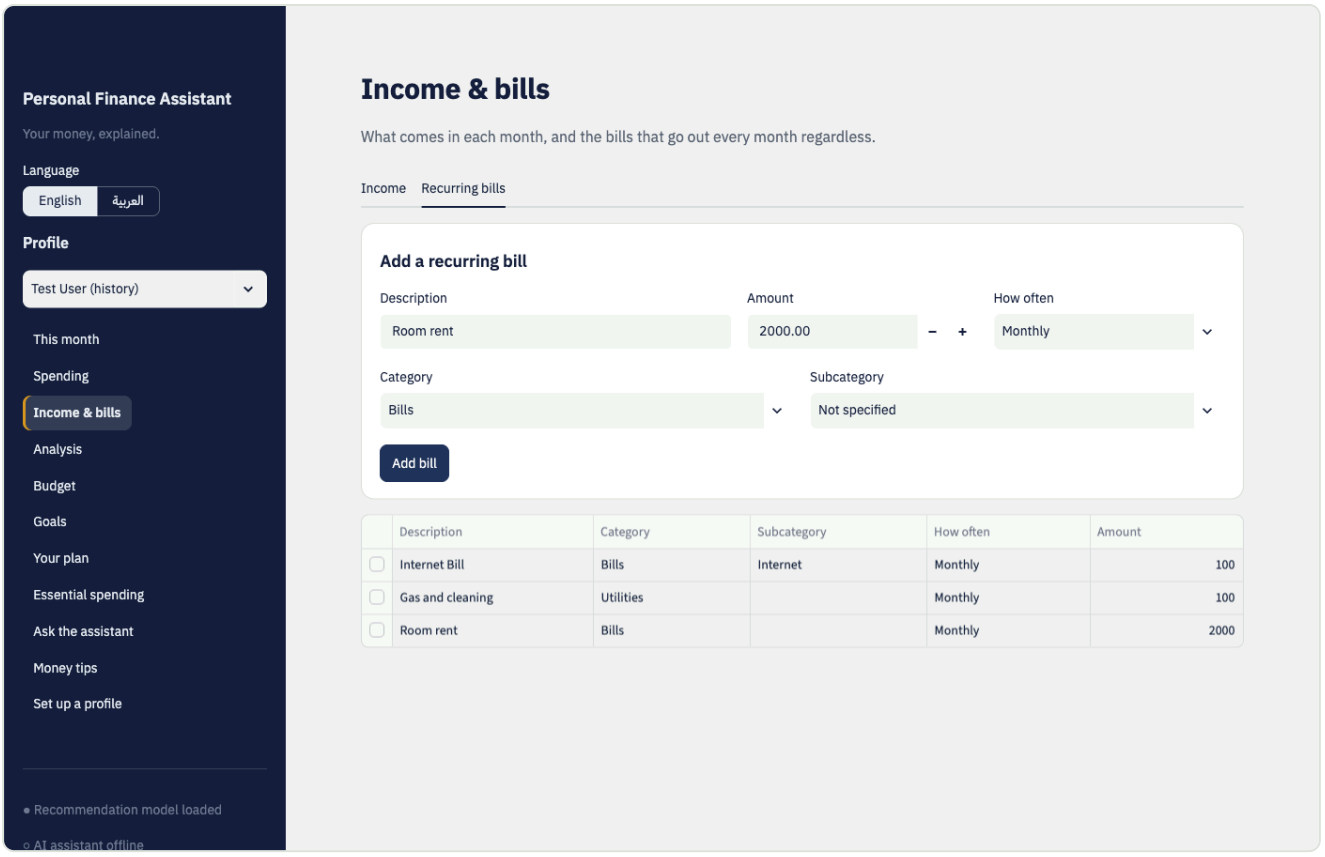



Income added

## UAT-09 Add a recurring bill

1. Open **Recurring bills** and enter "Room rent", 2,000, monthly, Bills.
2. Click **Add bill**.

**Expected:** the bill is saved and used first in the budget. **Actual:** saved with category Bills, 2,000 SDG, monthly. Pass. This screenshot was taken before defect D-03 was fixed; the form now clears.




Recurring bill added

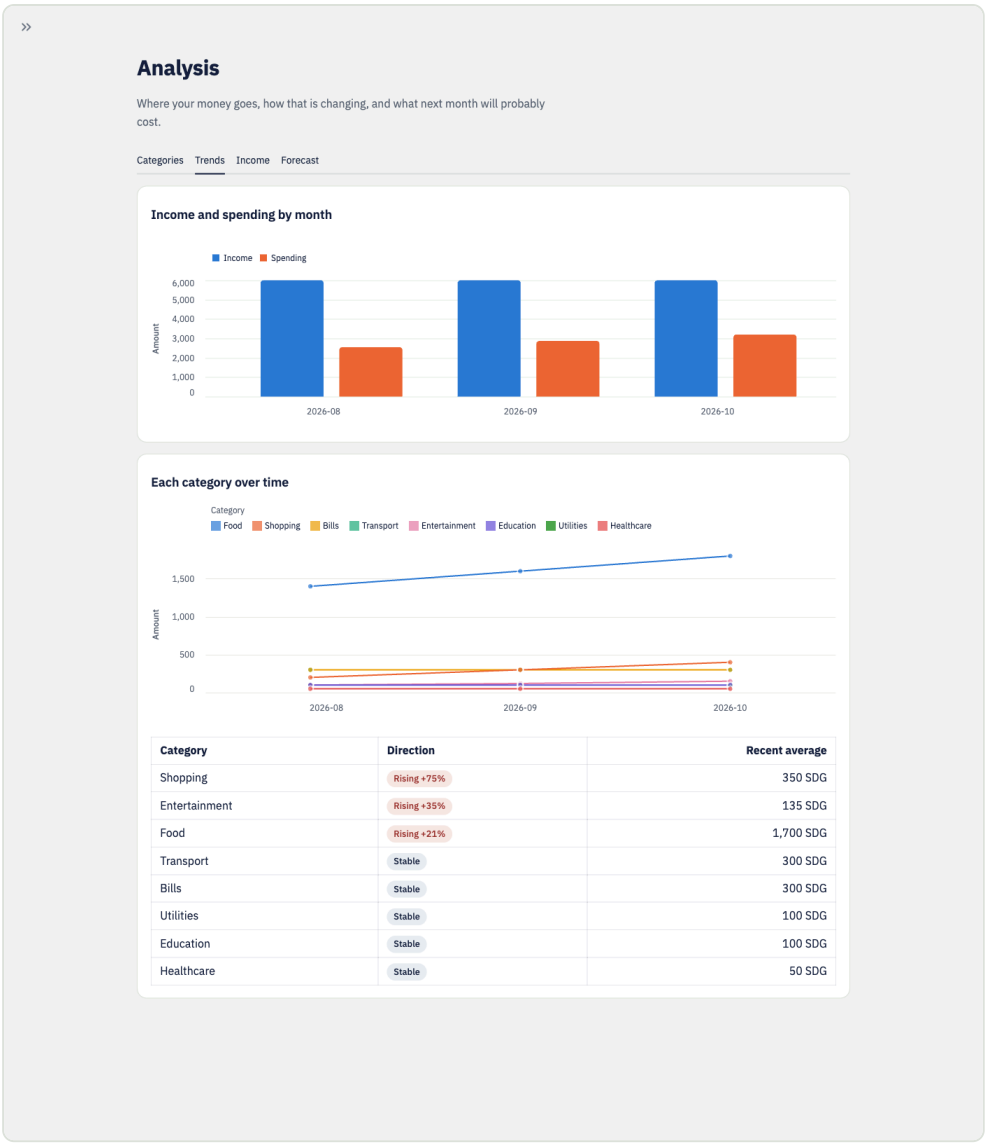
## UAT-10 Analysis

1. Open **Analysis** and go through the four tabs.

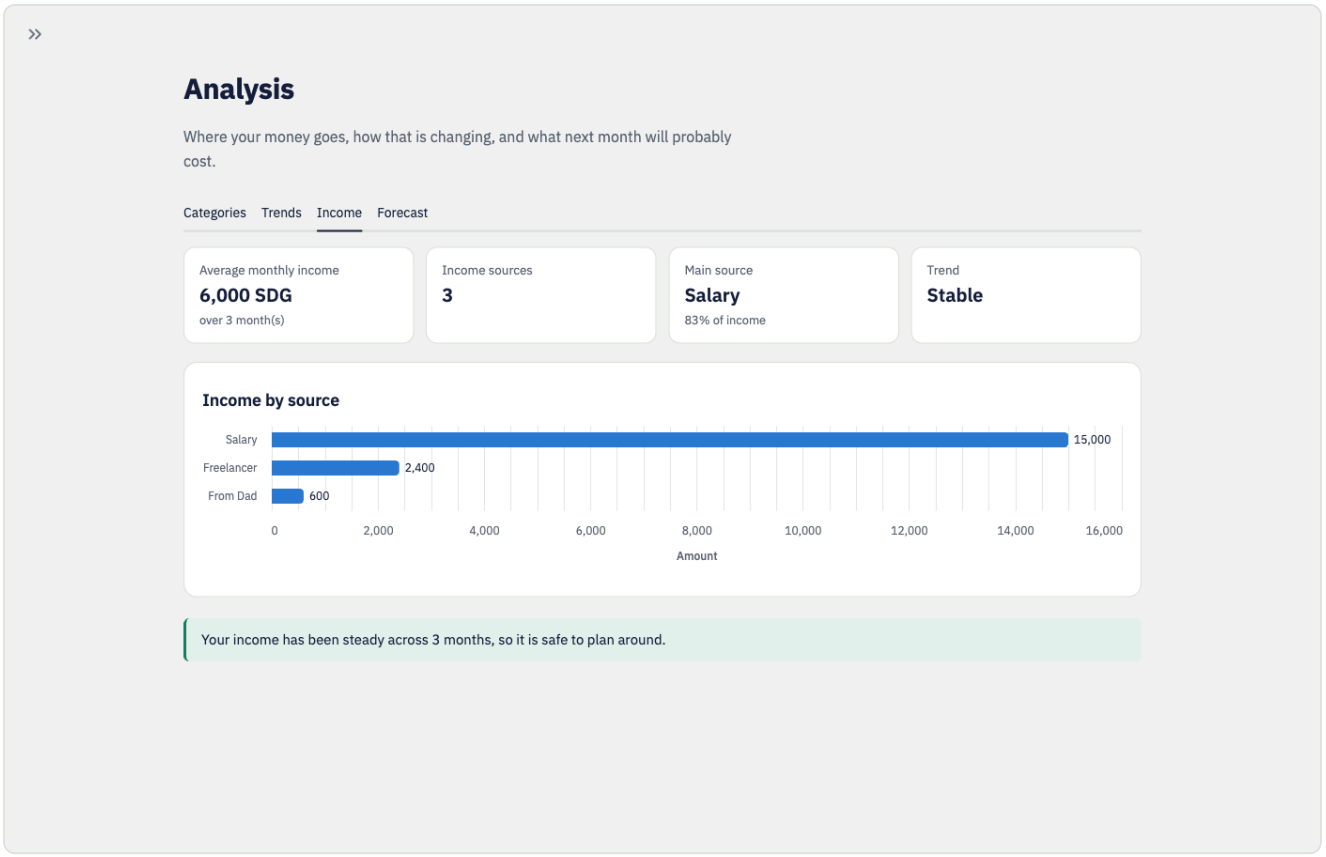
**Expected:** spending by category and inside a category; trends; income stability; next month's forecast.
**Actual:** Food is the largest category at 1,800 SDG. Food, Shopping and Entertainment are marked rising. Income is shown as steady over 3 months. The forecast is 3,523 SDG for next month. Pass.


Analysis
Where your money goes, how that is changing, and what next month will probably cost.
Categories
Trends
Income
Forecast
Showing 2026-10
Spending by category
Food
Shopping
Transport
Bills
Entertainment
Utilities
Education
Healthcare
1,800
400
300
300
150
100
100
50
0
200
400
600
800
1,000
1,200
1,400
1,600
1,800
Amount
Inside a category
Category
Food
Restaurant
Fast Food
Lunch
Dinner
Breakfast
800
400
300
200
100
0
50
100
150
200
250
300
350
400
450
500
550
600
650
700
750
800
Amount

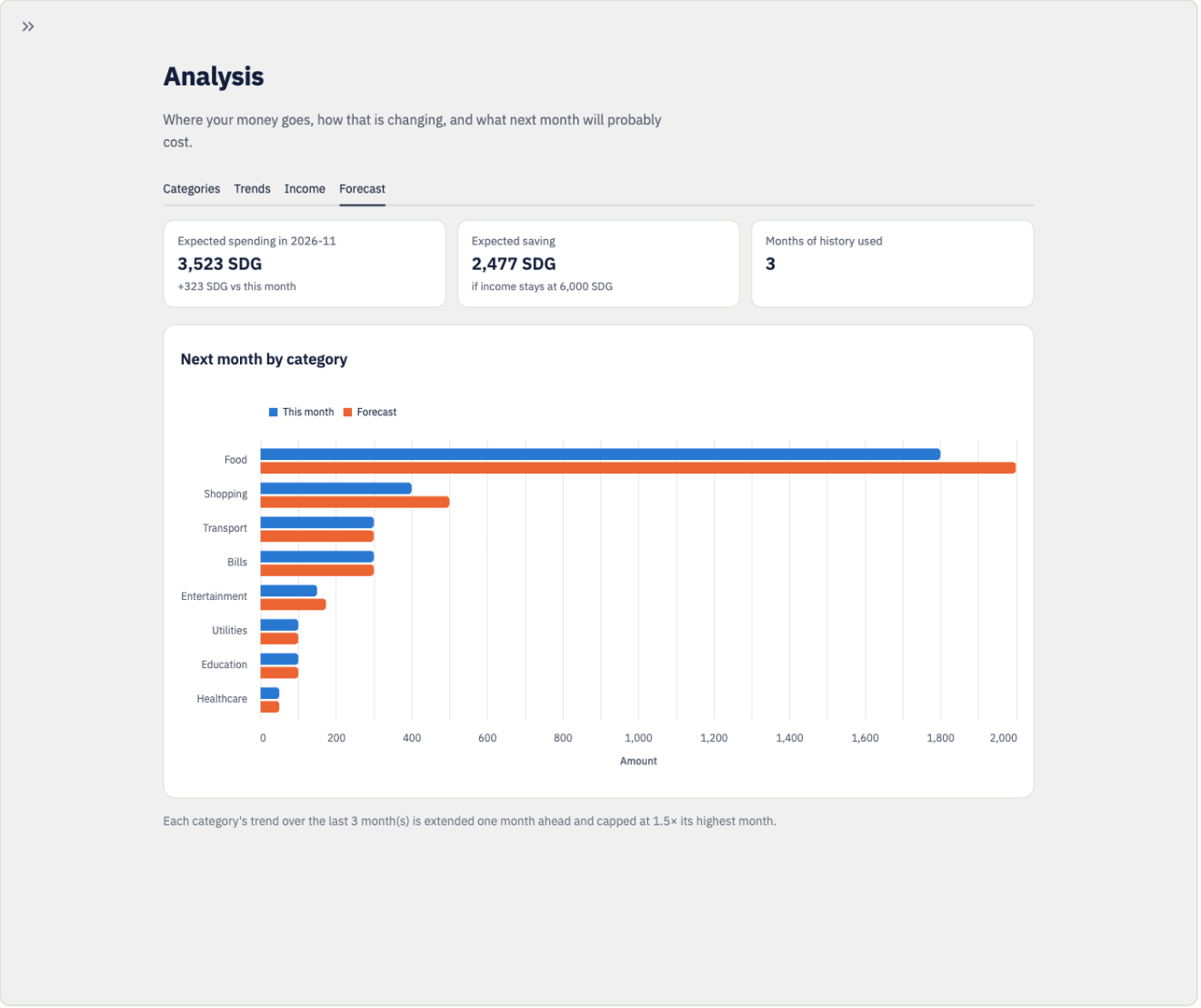
Categories


Analysis
Where your money goes, how that is changing, and what next month will probably cost.
Categories
Trends
Income
Forecast
Income and spending by month
Income
Spending
6,000
5,000
4,000
3,000
2,000
1,000
0
Amount
2026-08
2026-09
2026-10
Each category over time
Category
Food
Shopping
Bills
Transport
Entertainment
Education
Utilities
Healthcare
1,500
1,000
500
0
Amount
2026-08
2026-09
2026-10
Category
Direction
Recent average
Shopping
Rising +75%
350 SDG
Entertainment
Rising +35%
135 SDG
Food
Rising +21%
1,700 SDG
Transport
Stable
300 SDG
Bills
Stable
300 SDG
Utilities
Stable
100 SDG
Education
Stable
100 SDG
Healthcare
Stable
50 SDG

Trends


Analysis
Where your money goes, how that is changing, and what next month will probably cost.
Categories
Trends
Income
Forecast
Average monthly income
6,000 SDG
over 3 month(s)
Income sources
3
Main source
Salary
83% of income
Trend
Stable
Income by source
Salary
Freelancer
From Dad
15,000
2,400
600
0
2,000
4,000
6,000
8,000
10,000
12,000
14,000
16,000
Amount
Your income has been steady across 3 months, so it is safe to plan around.

Income

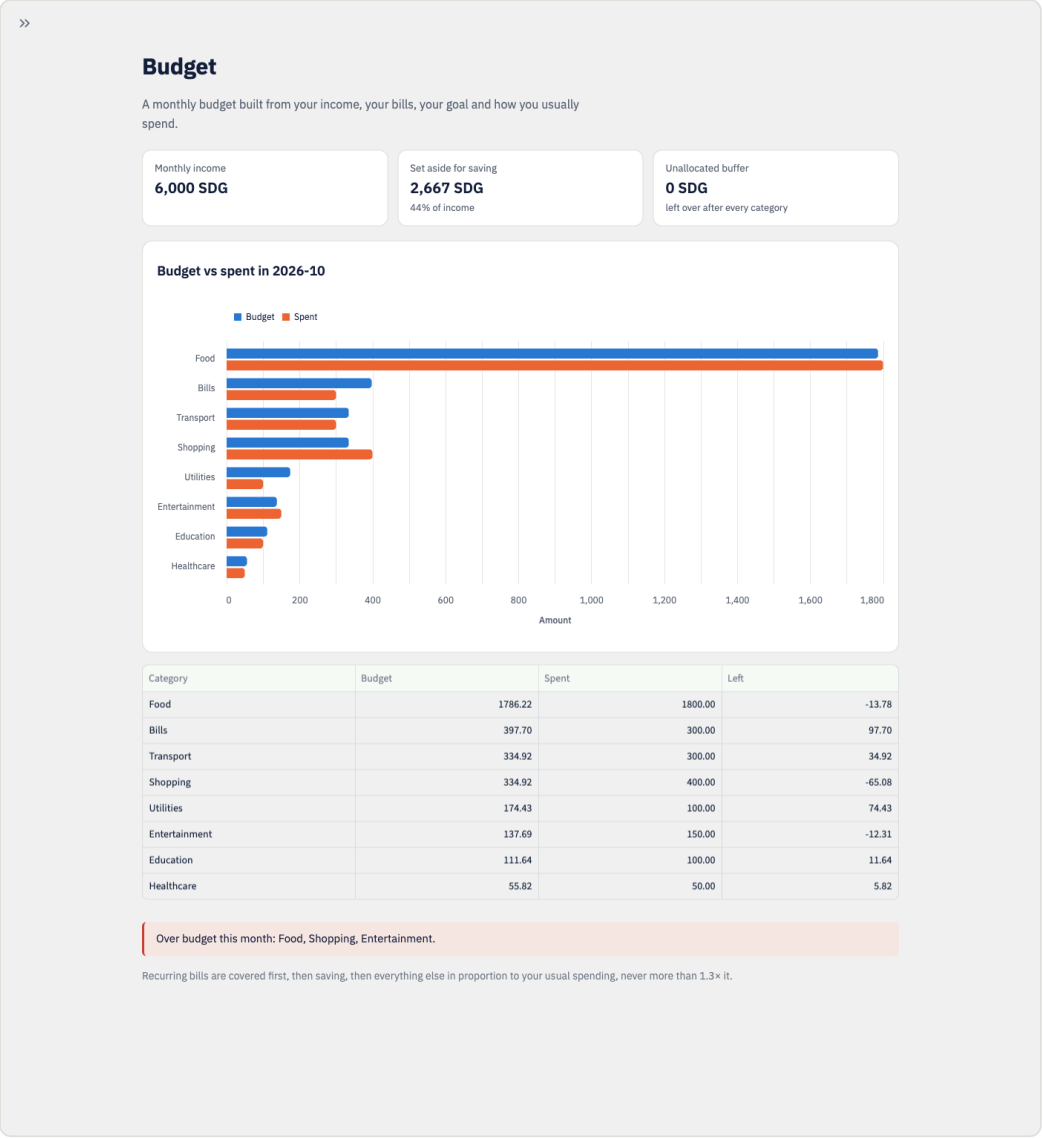



Forecast

## UAT-11 Budget

1. Open **Budget**.

**Expected:** a budget that never exceeds income, compared with this month's spending. **Actual:** income 6,000 SDG, 2,667 SDG set aside for saving, and a budget vs spent chart per category. Pass.

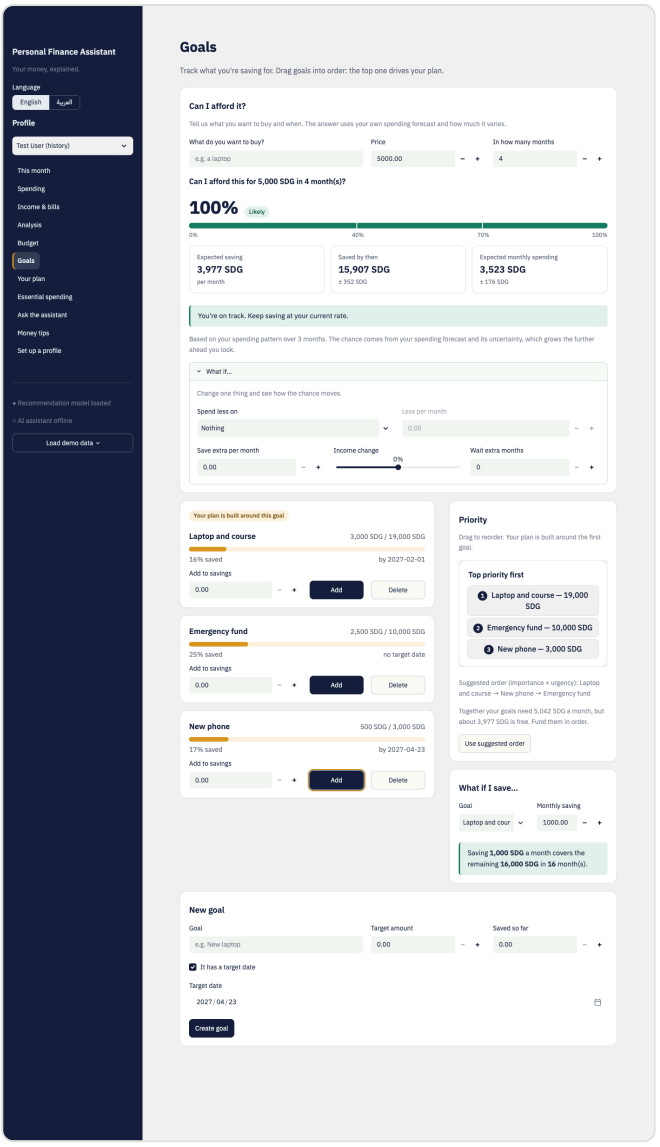



Budget vs spent

## UAT-12 Goals: create, add savings, delete

1. In **New goal**, enter "New phone" and 3,000, then click **Create goal**.
2. On its card, add 500 and click **Add**.
3. Click **Delete** on the card.

**Expected:** the goal appears, shows 500 / 3,000 SDG, and then disappears. **Actual:** as expected, checked through the API at each step. The form and the savings field clear after each save. Pass.

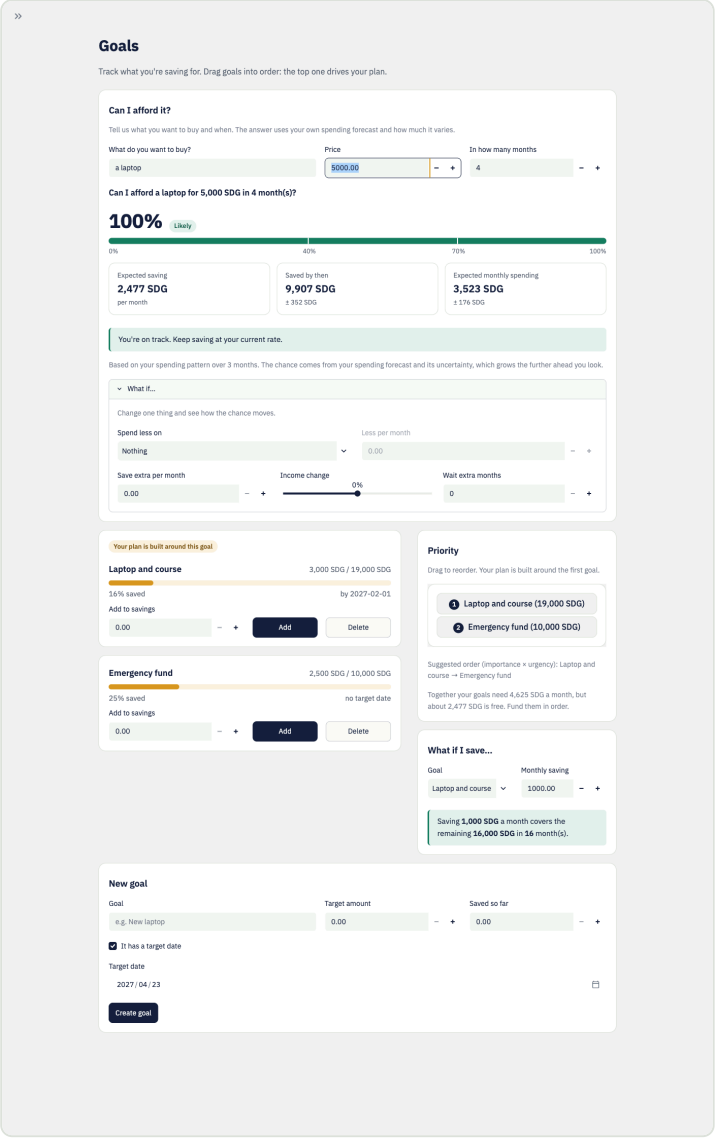



Goal created and funded

## UAT-13 "Can I afford it?"

1. Open **Goals**. Under **Can I afford it?** enter "a laptop", 5,000 SDG, 4 months.

**Expected:** a probability, a label, the expected saving, savings by then with a range, and an action.
**Actual:** 100%, Likely. Expected saving is 2,477 SDG a month, and savings by then 9,907 ± 352 SDG. The answer is labelled "Based on your spending pattern over 3 months". Pass.

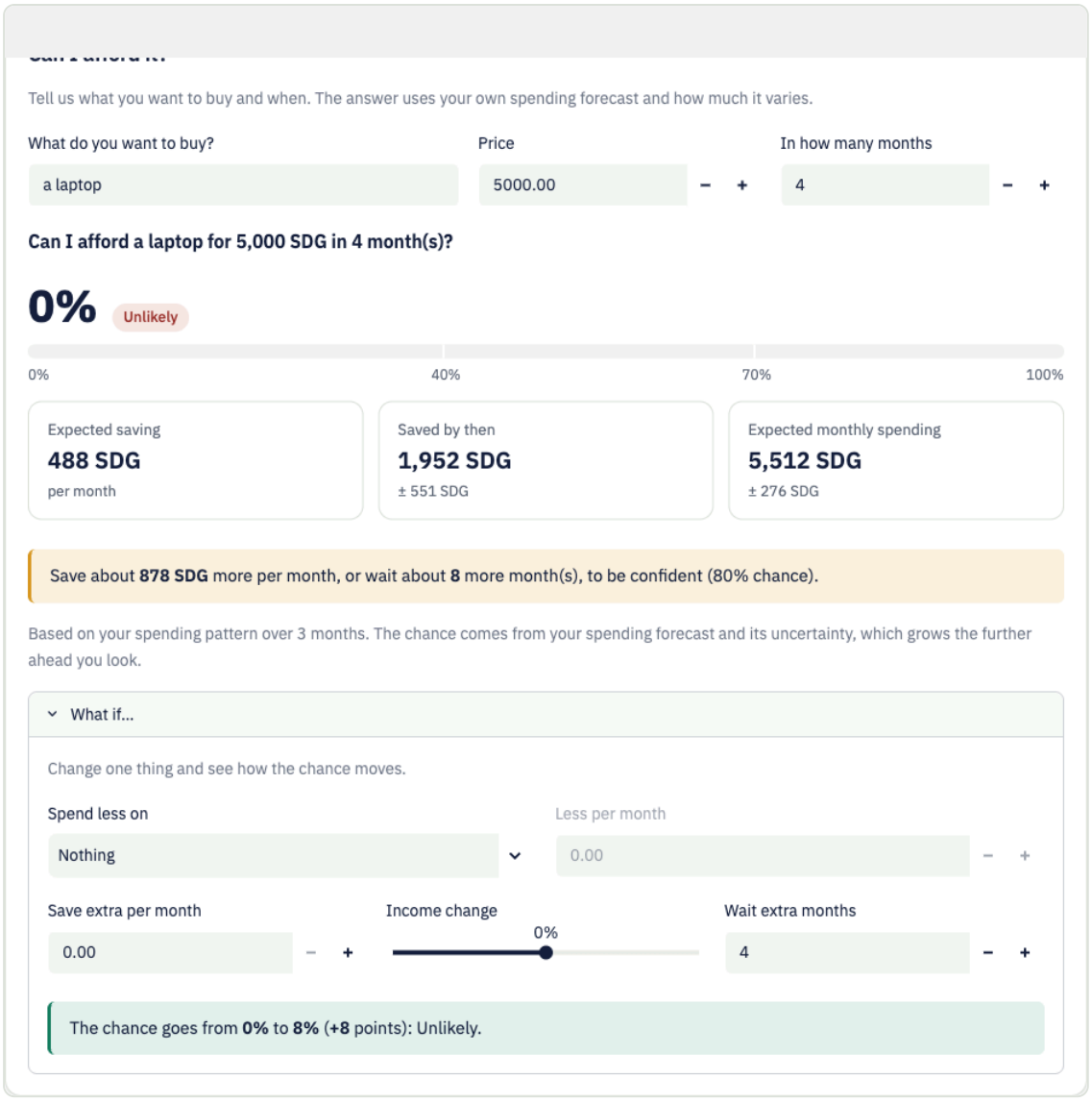



Can I afford it

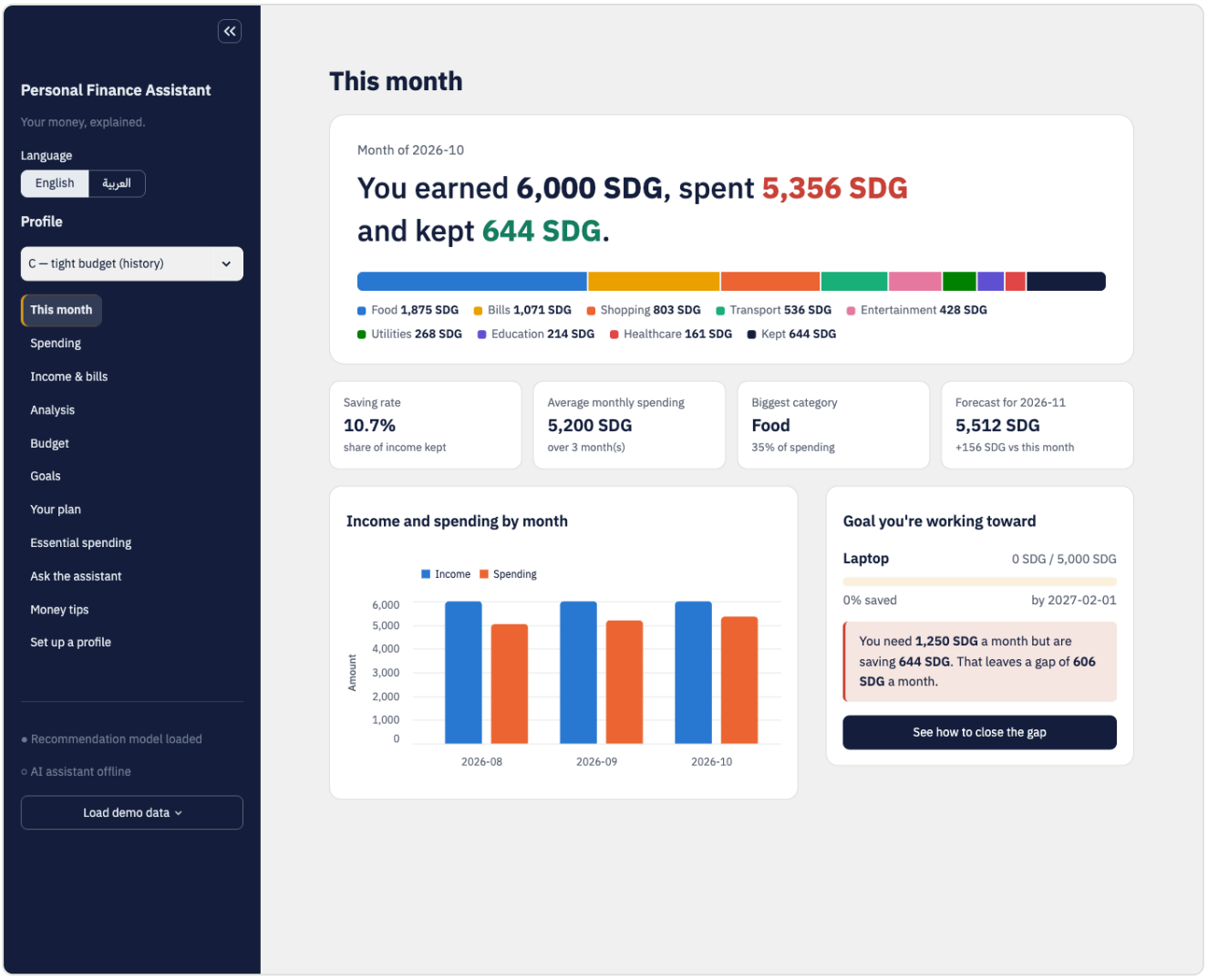
## UAT-14 Personalization and what-if

1. Switch **Profile** to *C — tight budget* and ask the same question.
2. Under **What if...**, set **Wait extra months** to 4.

**Expected:** a different answer for different money, and a recalculated chance. **Actual:** 0%, Unlikely. The action reads "Save about 878 SDG more per month, or wait about 8 more months". The what-if shows "The chance goes from 0% to 8% (+8 points)". Pass.




Profile C with what-if

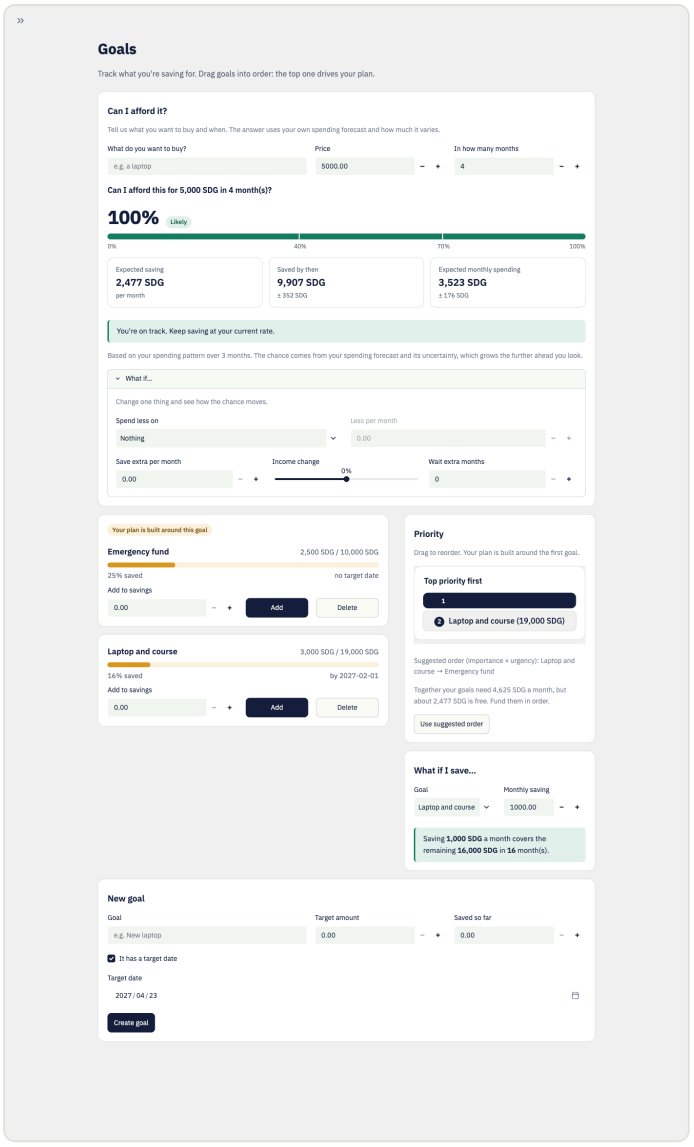



Profile C dashboard

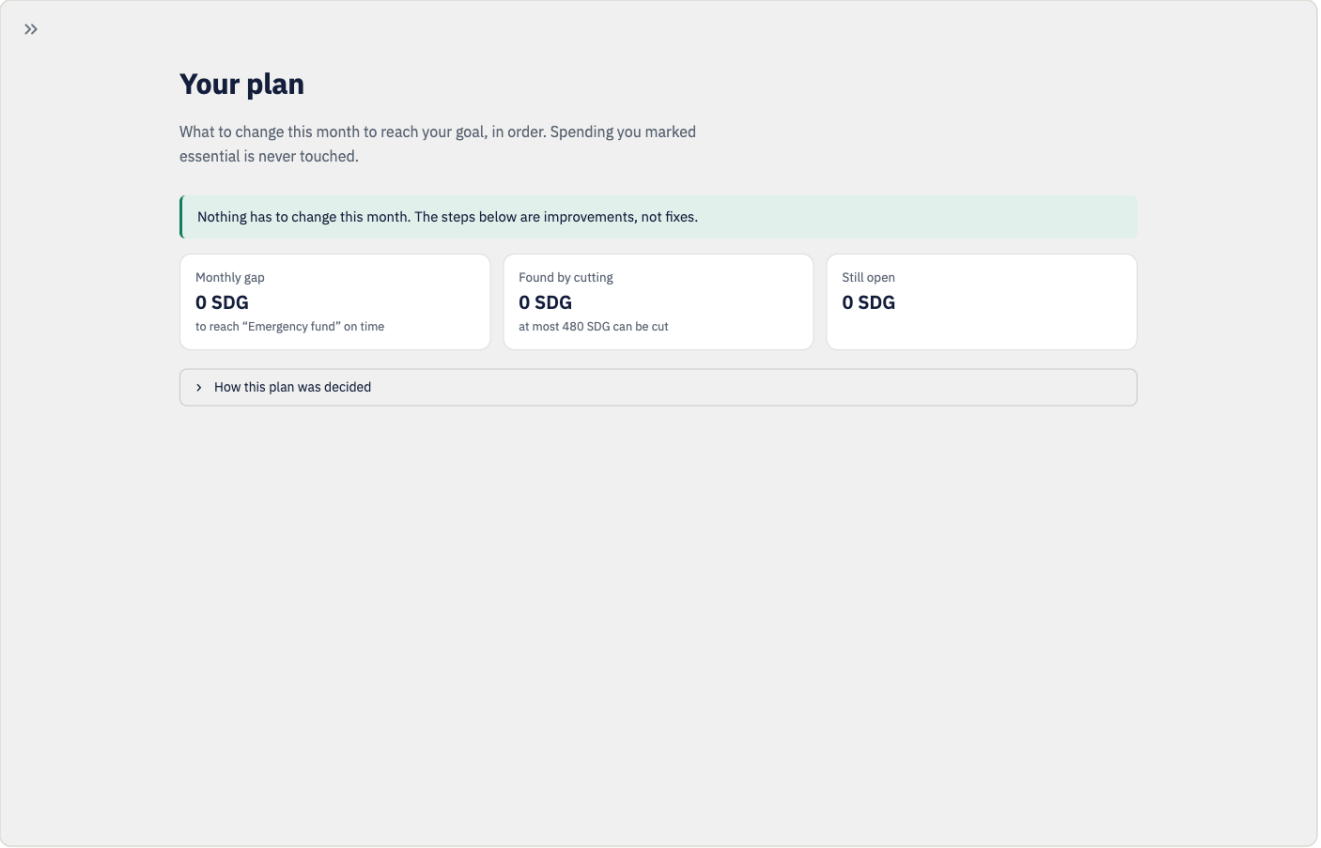
## UAT-15 Goal priority board

1. On **Goals → Priority**, drag *Emergency fund* above *Laptop and course*.
2. Open **Your plan**.

**Expected:** the order is saved, and the plan follows the new top goal. **Actual:** the order is saved (Emergency fund priority 1). The plan now targets Emergency fund and reads "No funding gap this month". Pass, after defect D-01 was fixed.




Priority after dragging




Plan after reordering

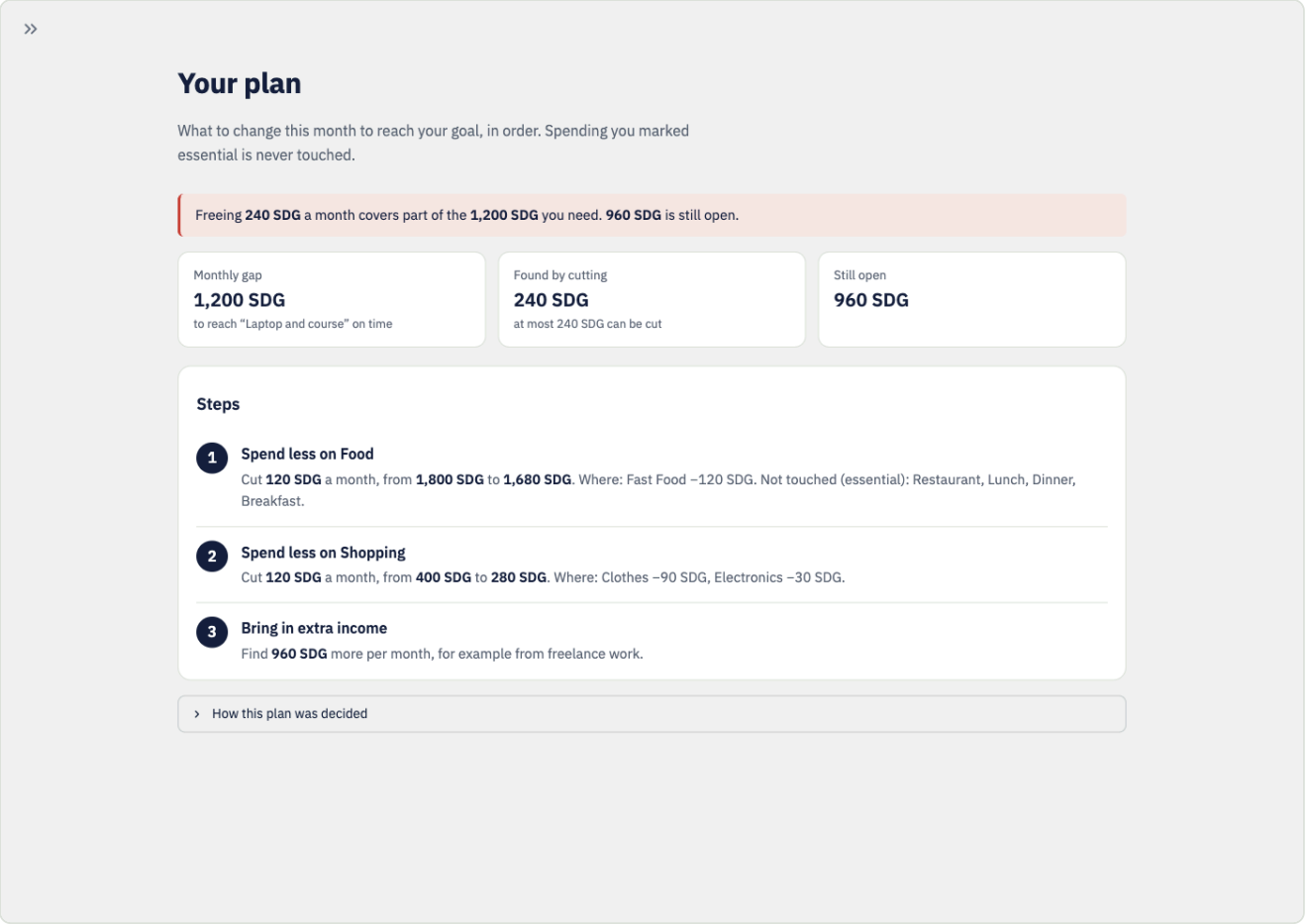
## UAT-16 Essential spending board

1. Open **Essential spending** and drag *Food › Restaurant* into **Essential: never cut**.
2. Open **Your plan**.

**Expected:** the preference is saved and the plan stops cutting Restaurant, with no retraining. **Actual:** saved as essential. The Food cut fell from 360 to 120 SDG, now taken only from Fast Food. Pass.




Restaurant moved to Essential

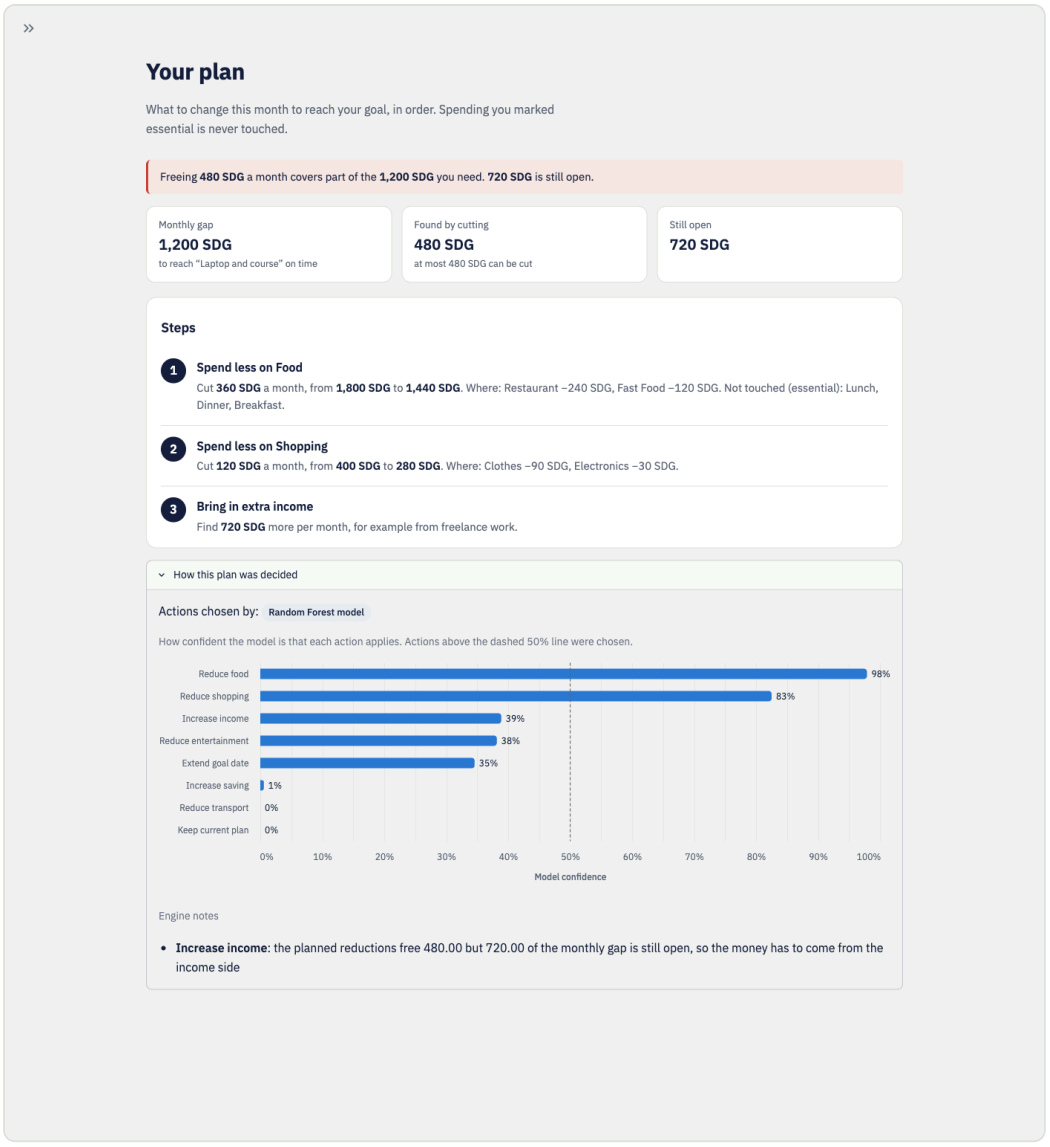



Plan after the change

## UAT-17 The plan

1. Reload demo data and open **Your plan**, then **How this plan was decided**.

**Expected:** numbered steps with amounts, essential items marked as not touched, and the model's confidence. **Actual:** the steps are "cut 360 SDG from Food (Restaurant −240, Fast Food −120; Lunch, Dinner, Breakfast not touched)", "cut 120 SDG from Shopping" and "find 720 SDG more per month". The model is 98% confident about reducing food and 83% about reducing shopping. Pass.

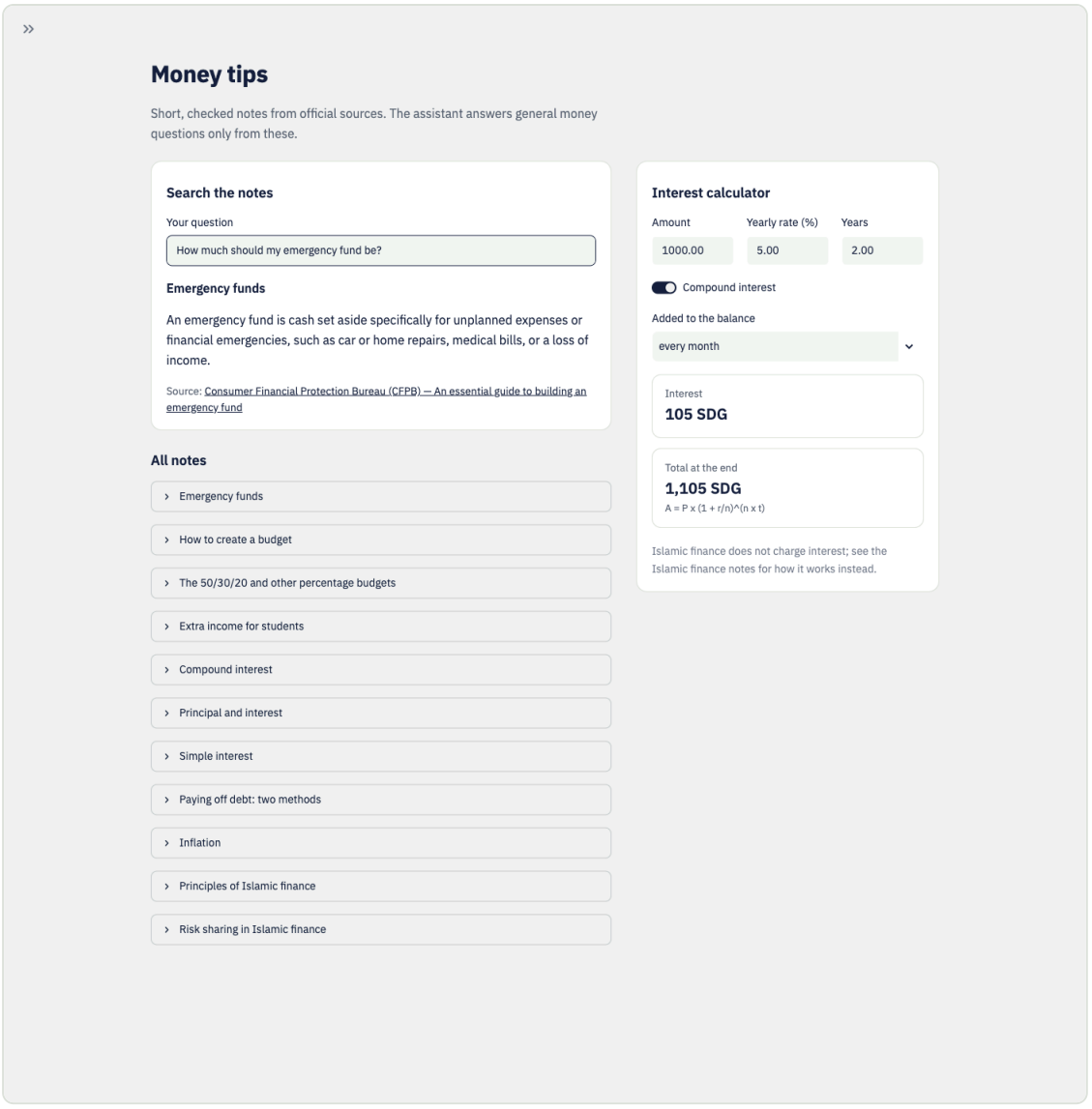



Plan

## UAT-18 Money tips

1. Open **Money tips** and search "How much should my emergency fund be?".
2. Search "What will the dollar rate be next month?".

**Expected:** the first finds a note with its source; the second finds nothing. **Actual:** the first returns the CFPB note on emergency funds with a link. The second returns "None of the trusted notes cover this". Pass.

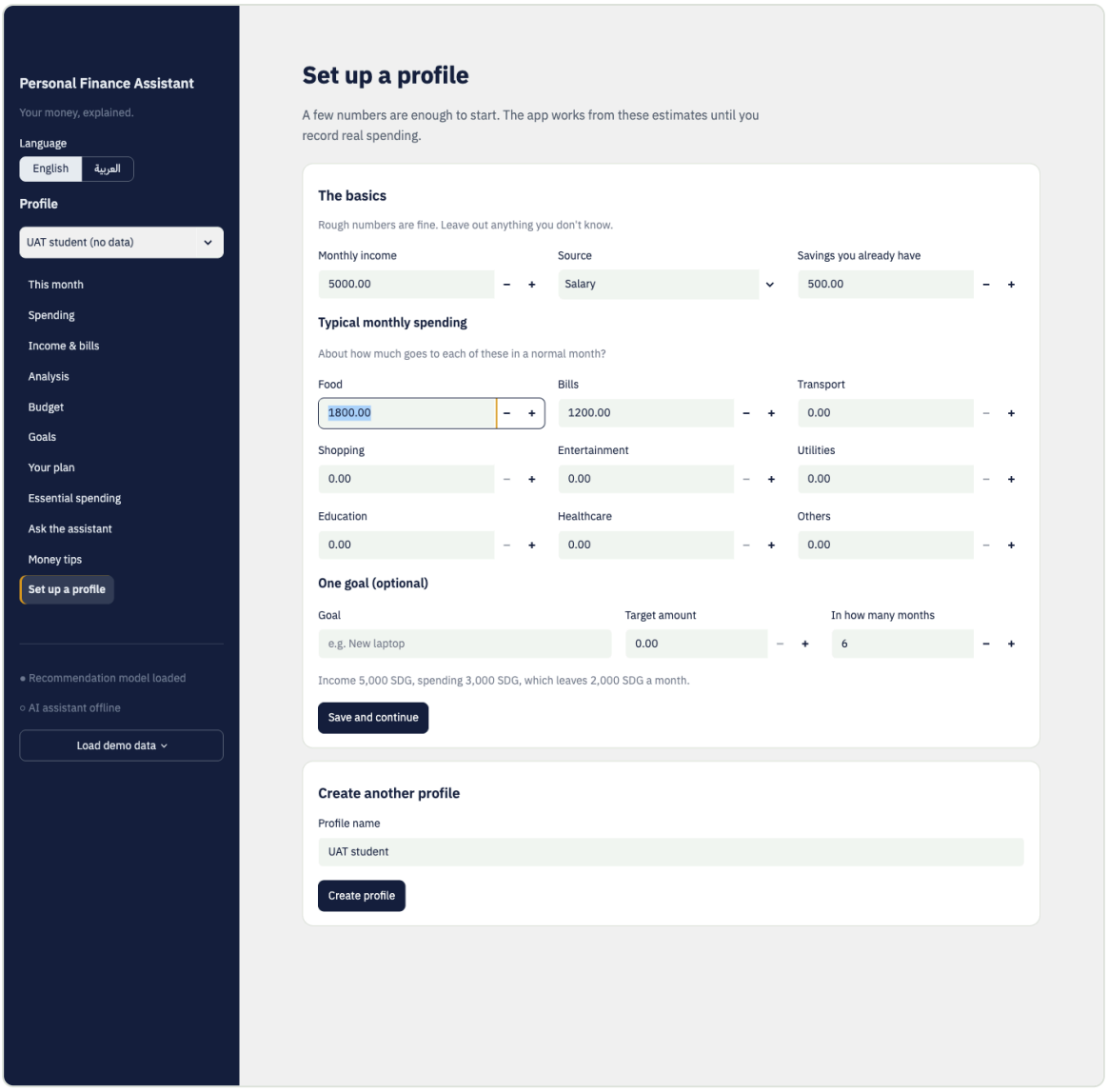



Money tips search

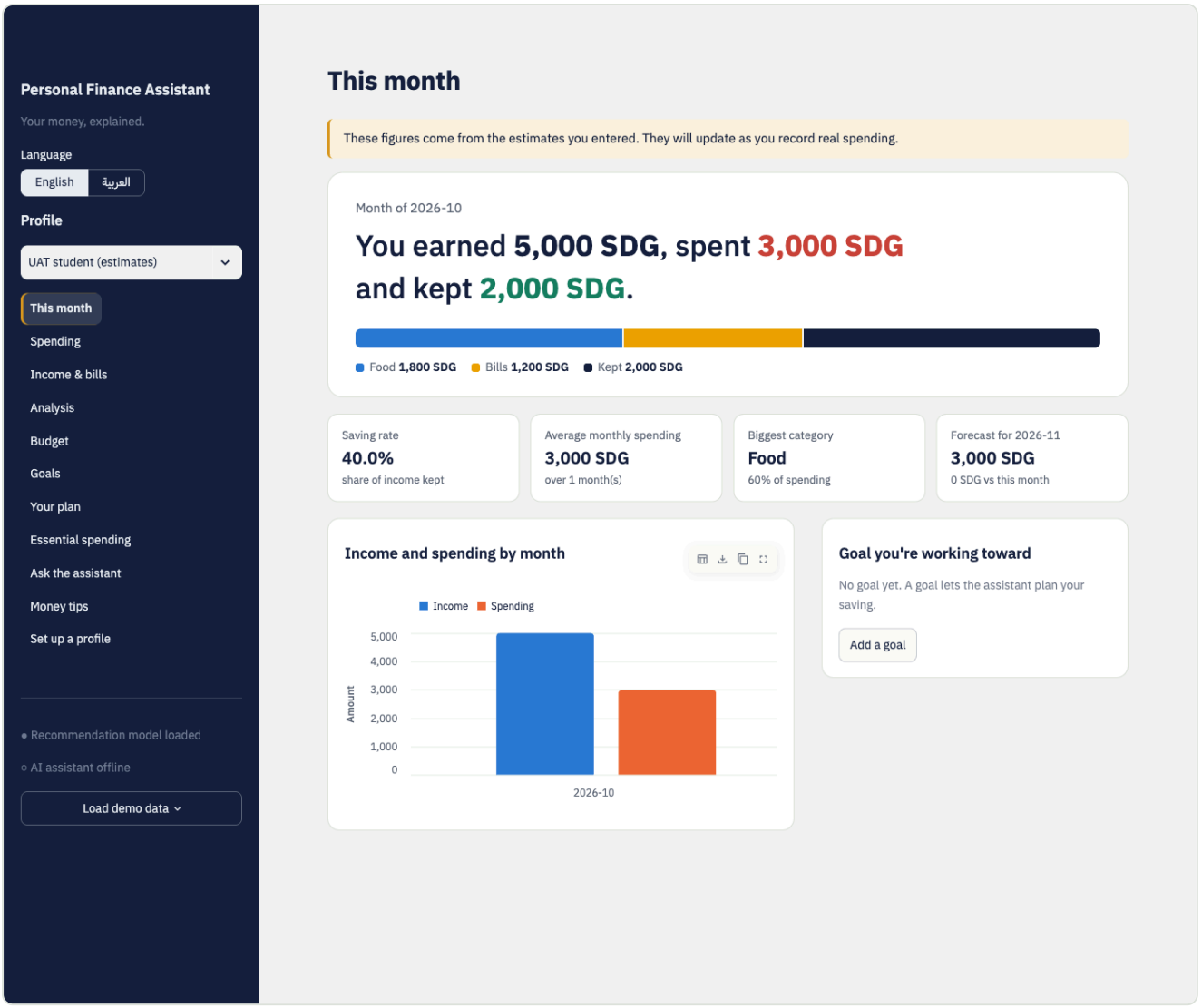
## UAT-19 Set up a new profile

1. Open **Set up a profile**, type "UAT student" and click **Create profile**.
2. Enter income 5,000, Food 1,800, Bills 1,200 and savings 500, then click **Save and continue**.

**Expected:** a new profile, its estimates saved, and the dashboard opened for it. **Actual:** the profile is created and selected, and its data state is "entered". The dashboard reads "You earned 5,000 SDG, spent 3,000 SDG and kept 2,000 SDG", under the note "These figures come from the estimates you entered". Pass.




Setup form filled

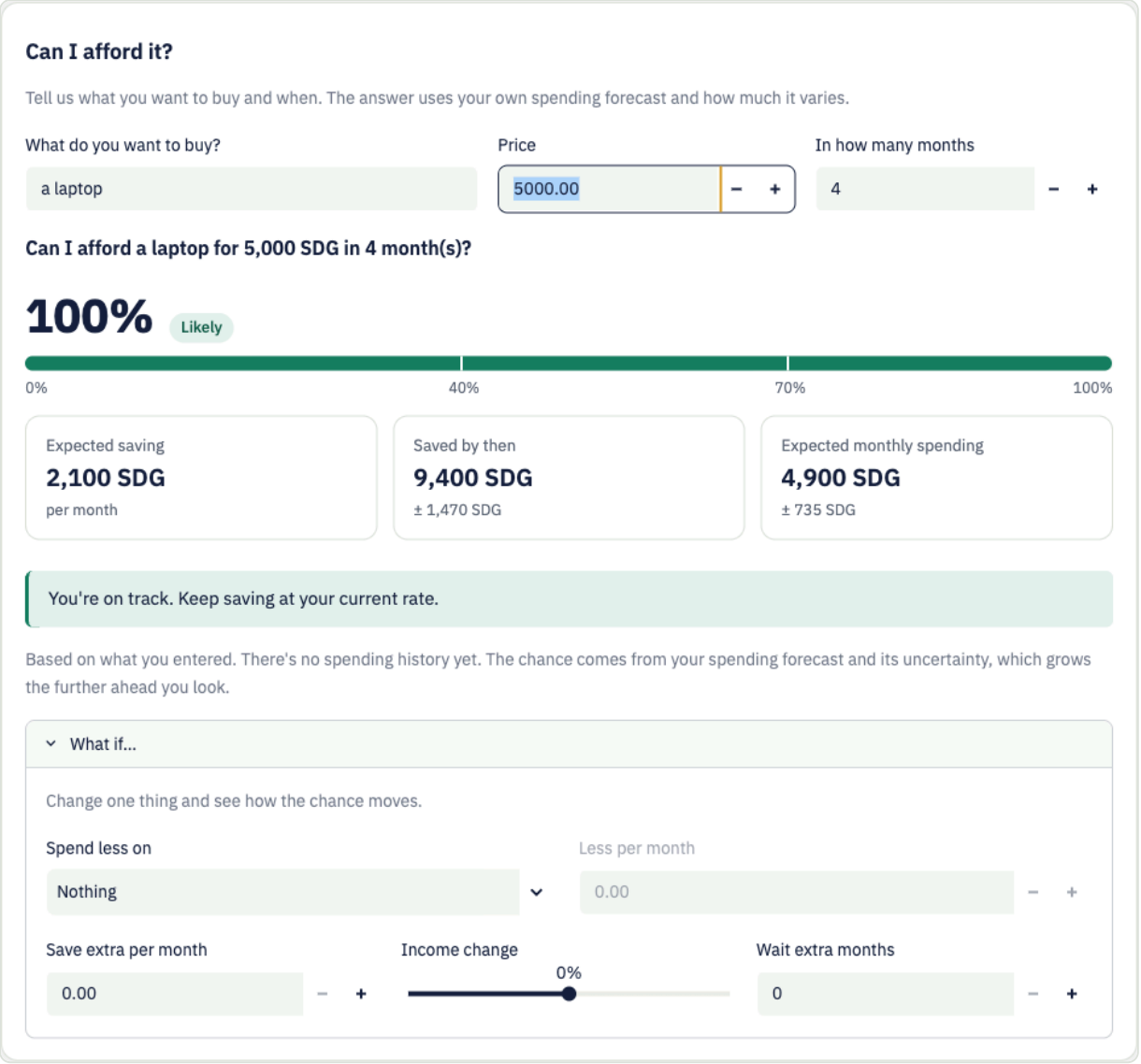



New profile dashboard

## UAT-20 Honest answers for a new user

1. Select *New user — onboarding only* and ask "Can I afford it?" for a 5,000 laptop in 4 months.

**Expected:** the answer says it is based on what the user entered, not on history. **Actual:** "Based on what you entered. There's no spending history yet." The uncertainty is ±15% (±735 SDG). Pass.




New user assessment

## UAT-21 Profile kept across a language switch

1. Select profile C, switch to العربية, change page, then return to the dashboard. Repeat 3 times with different timing.

**Expected:** profile C stays selected. **Actual:** profile C's figures (5,356 SDG spent) were shown in all 3 runs. Pass, after defect D-06 was fixed.

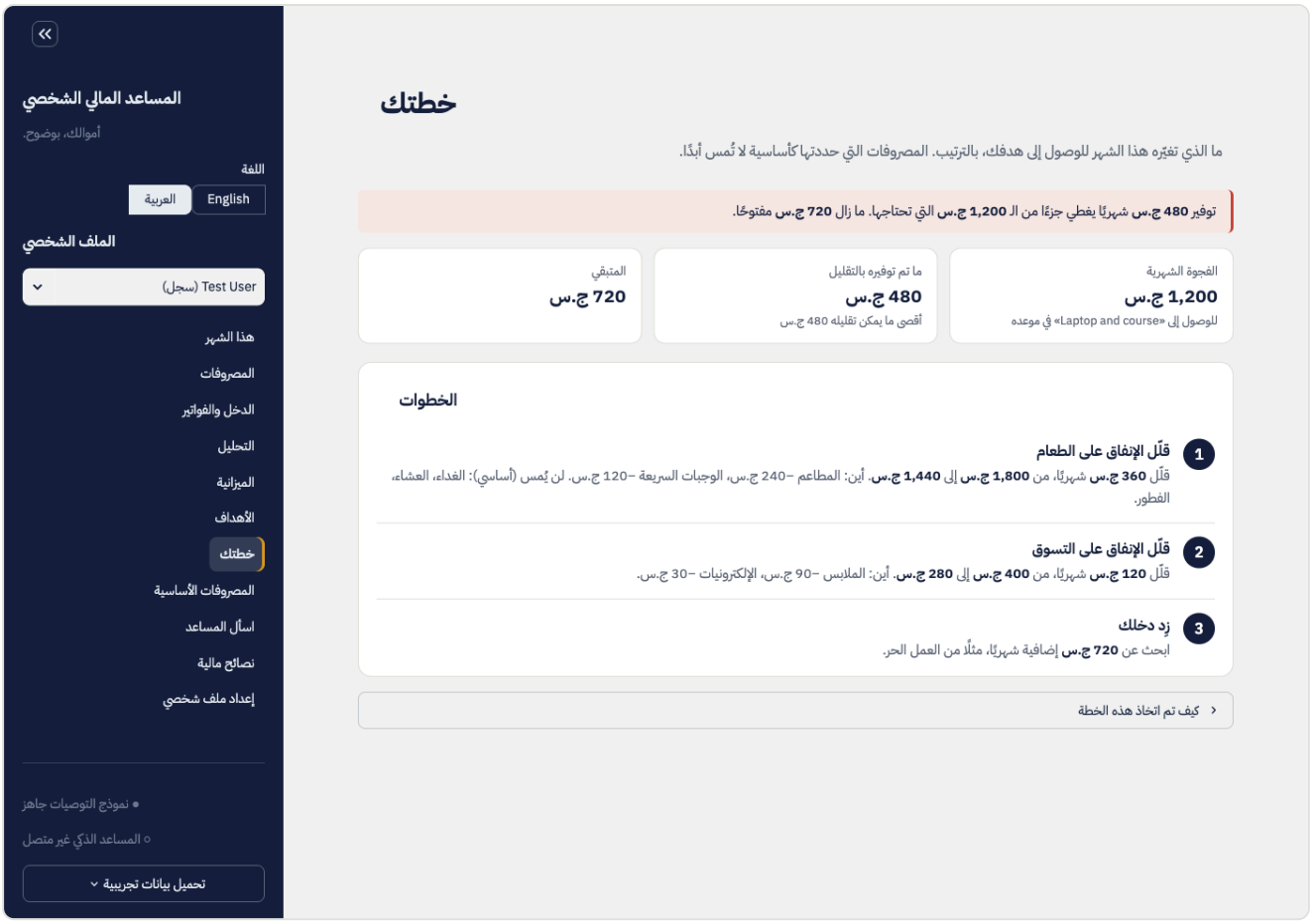
## UAT-22 Arabic

1. Switch the language to العربية and open the dashboard, the plan, essential spending and goals.

**Expected:** every text is in Arabic, the layout is right-to-left, and amounts read "ج.س". **Actual:** as expected, including category names, chart labels and the plan's sentences. English goal names inside the priority board display correctly after defect D-05 was fixed. Pass.



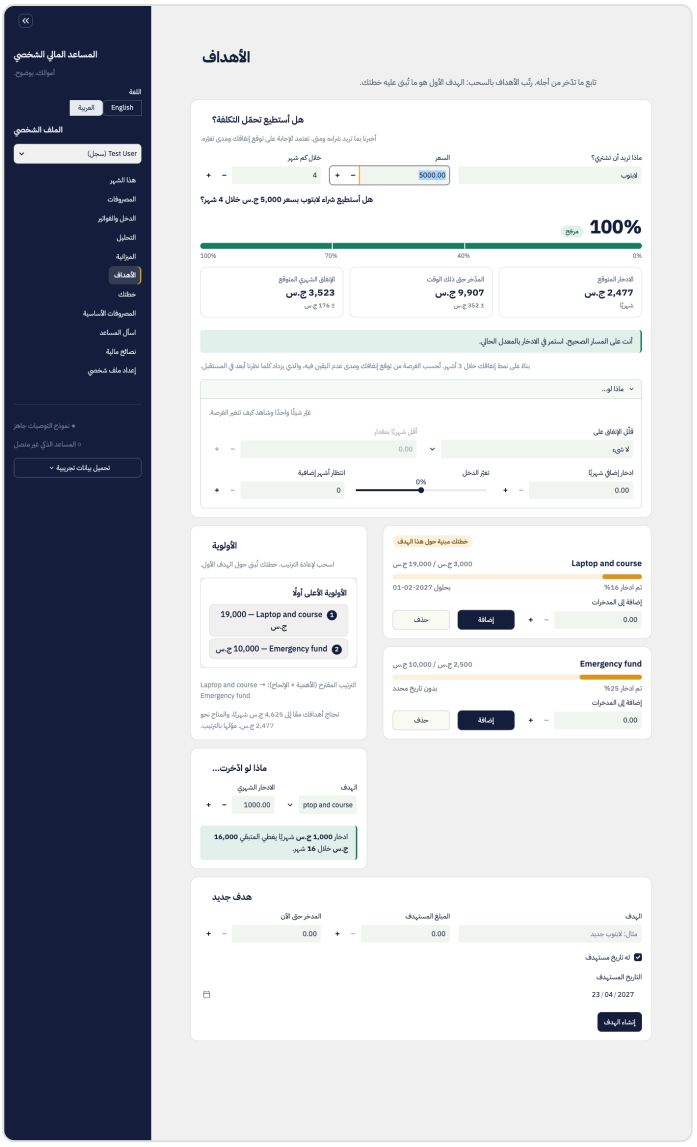

Arabic dashboard


المساعد المالي الشخصي
أموالك، بوضوح.
اللغة
English
العربية
الملف الشخصي
Test User (سجل)
هذا الشهر
المصروفات
الدخل والفواتير
التحليل
الميزانية
الأهداف
خطتك
المصروفات الأساسية
اسأل المساعد
نصائح مالية
إعداد ملف شخصي
نموذج التوصيات جاهز
المساعد الذكي غير متصل
تحميل بيانات تجريبية
خطتك
ما الذي تغيّره هذا الشهر للوصول إلى هدفك، بالترتيب. المصروفات التي حددتها كأساسية لا تُمس أبدًا.
توفير 480 ج.س شهريًا يغطي جزءًا من الـ 1,200 ج.س التي تحتاجها. ما زال 720 ج.س مفتوحًا.
الفجوة الشهرية
1,200 ج.س
للوصول إلى «Laptop and course» في موعده
ما تم توفيره بالتقليل
480 ج.س
أقصى ما يمكن تقليله 480 ج.س
المتبقي
720 ج.س
الخطوات
1
قلّل الإنفاق على الطعام
قلّل 360 ج.س شهريًا، من 1,800 ج.س إلى 1,440 ج.س. أين: المطاعم –240 ج.س، الوجبات السريعة –120 ج.س. لن يُمس (أساسي): الغداء، العشاء، الفطور.
2
قلّل الإنفاق على التسوق
قلّل 120 ج.س شهريًا، من 400 ج.س إلى 280 ج.س. أين: الملابس –90 ج.س، الإلكترونيات –30 ج.س.
3
زِد دخلك
ابحث عن 720 ج.س إضافية شهريًا، مثلًا من العمل الحر.
كيف تم اتخاذ هذه الخطة

Arabic plan




Arabic essential spending

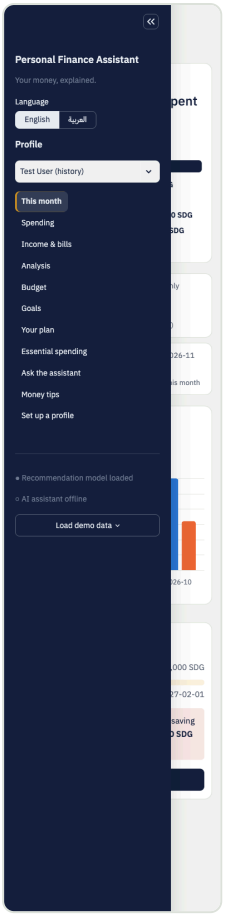
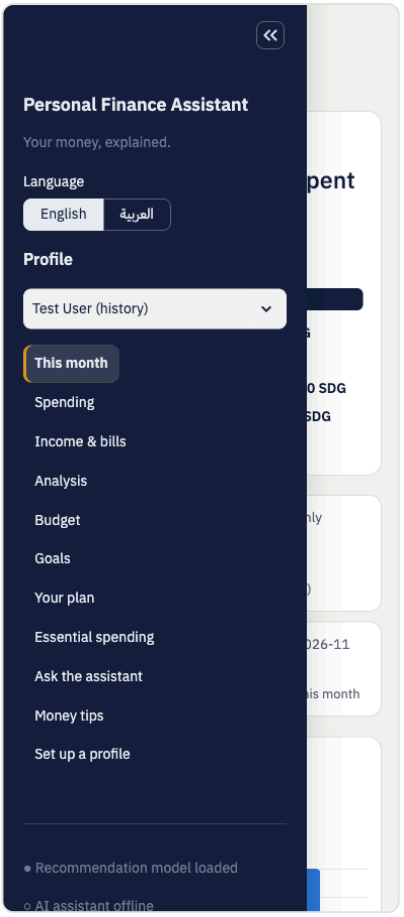
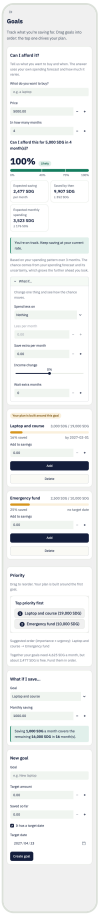



Arabic goals

## UAT-23 Mobile

1. Open the app at 390 px wide (a phone) and visit every page.

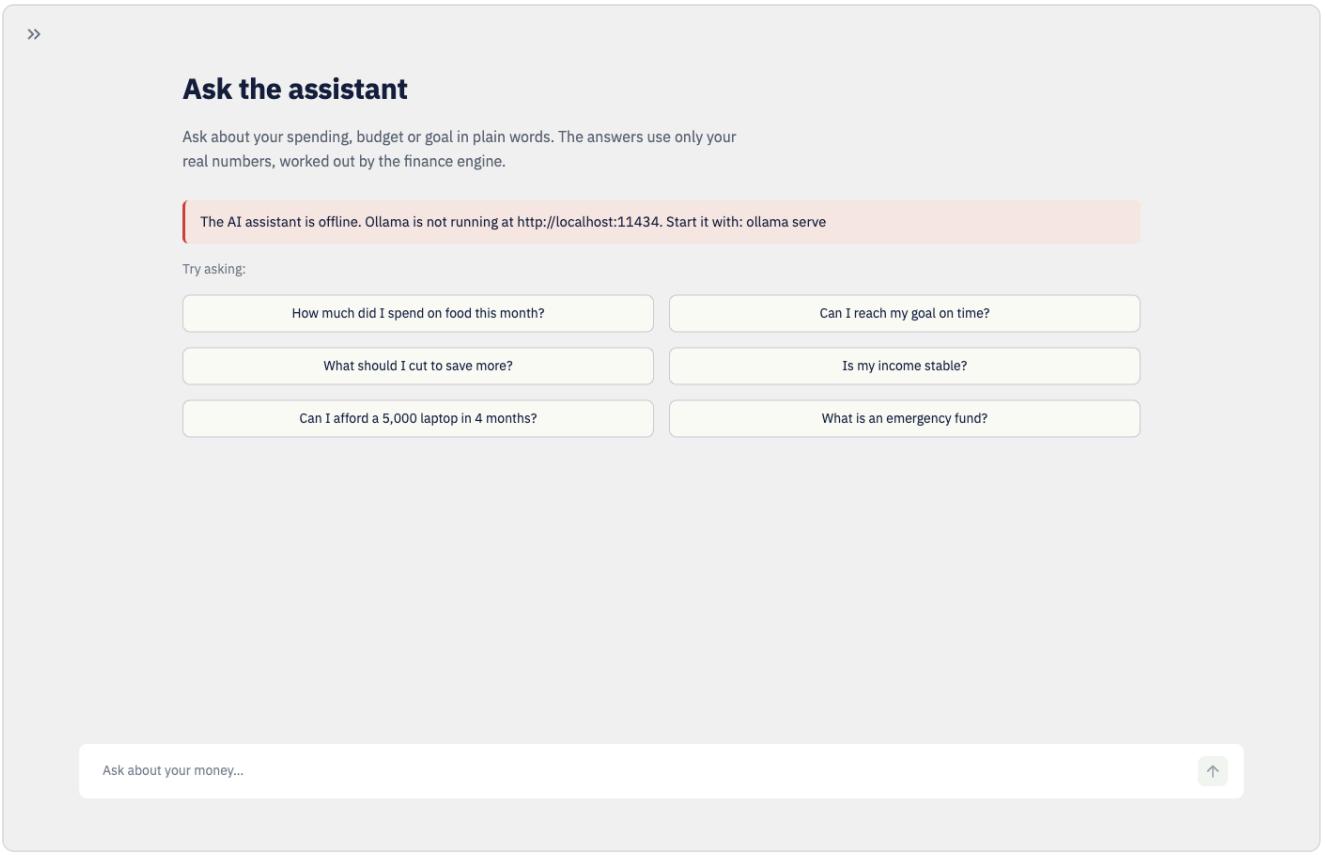
**Expected:** no sideways scrolling, a readable layout, and a working menu. **Actual:** sideways overflow is 0 px on every page tested. The sidebar folds behind a menu button, columns stack, and the essential-spending board stacks its two columns. Dragging at phone width was not tested. Pass.

| Dashboard | Menu | Goals |
| --- | --- | --- |
|  |  |  |
| Phone dashboard | Phone menu | Phone goals |

## UAT-24 Assistant offline

1. With Ollama not running, open **Ask the assistant**.

**Expected:** a clear message and the command to fix it; the rest of the app works. **Actual:** "The AI assistant is offline. Ollama is not running at http://localhost:11434. Start it with: ollama serve". The sidebar shows "AI assistant offline". Pass.

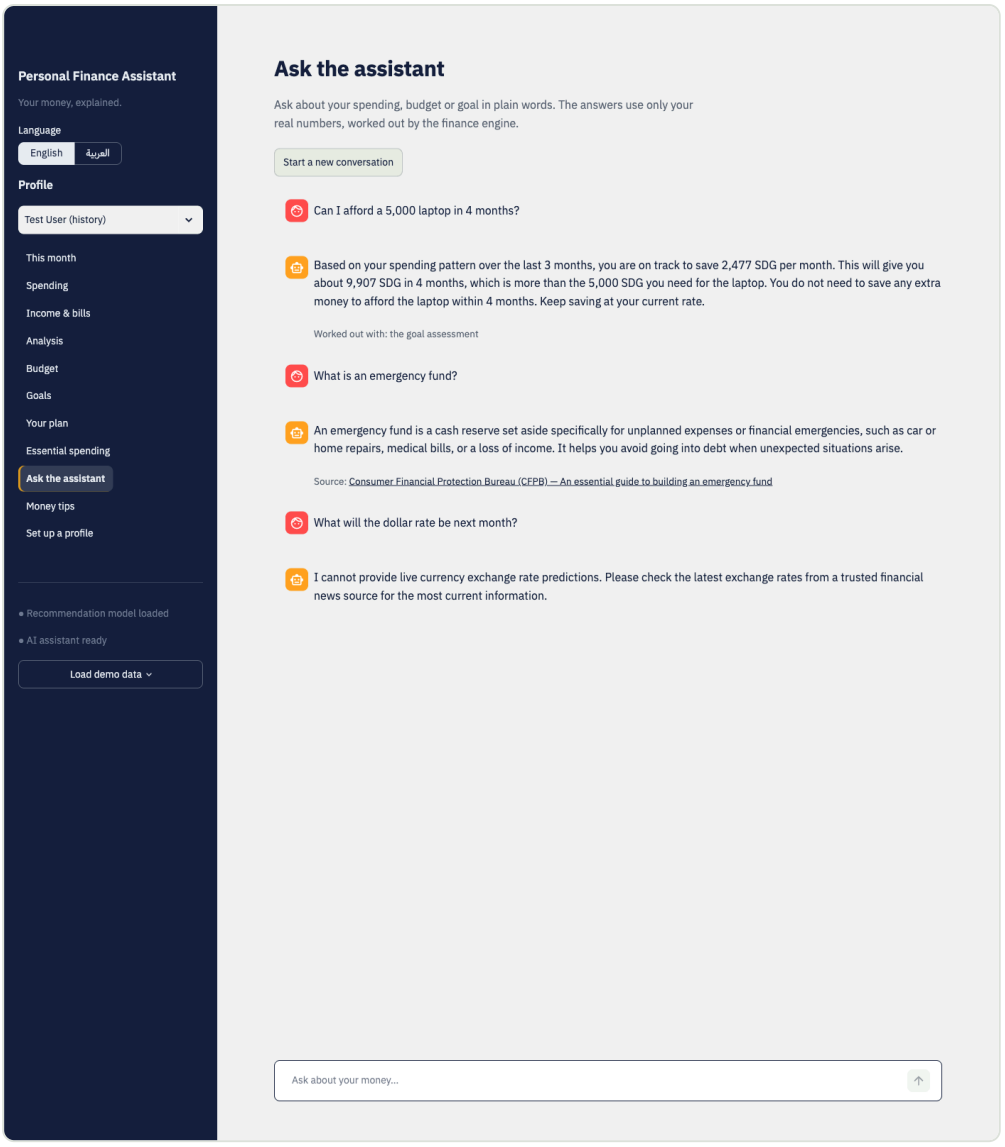



Assistant offline

## UAT-25 Assistant answers with its tools and sources

1. Install the model: `ollama serve`, `ollama pull qwen2.5:3b`.
2. Ask "Can I afford a 5,000 laptop in 4 months?" and "What is an emergency fund?".

**Expected:** numbers matching the Goals page and the tool named under the answer; a general answer taken from a verified note, with its source shown. **Actual:** "...you are on track to save 2,477 SDG per month. This will give you about 9,907 SDG in 4 months, which is more than the 5,000 SDG you need...", with "Worked out with: the goal assessment". The emergency-fund answer follows the CFPB note and shows "Source: Consumer Financial Protection Bureau (CFPB) — An essential guide to building an emergency fund". Pass, after defects D-08, D-10, D-13 and D-14 were fixed. Before those fixes, this case failed: the answer was in dollars, then a general answer came from the model's memory with an invented source, then the laptop answer contradicted itself.




Assistant answers in English

## UAT-26 A live-data question is refused

1. Ask "What will the dollar rate be next month?".

**Expected:** the assistant says it cannot know live data. **Actual:** "I cannot provide live currency exchange rate predictions. Please check the latest exchange rates from a trusted financial news source..." Pass.

## UAT-27 Currency

**Expected:** every amount in Sudanese pounds. **Actual:** every screenshot above shows SDG in English and ج.س in Arabic, and the assistant is told to answer in Sudanese pounds. Pass.

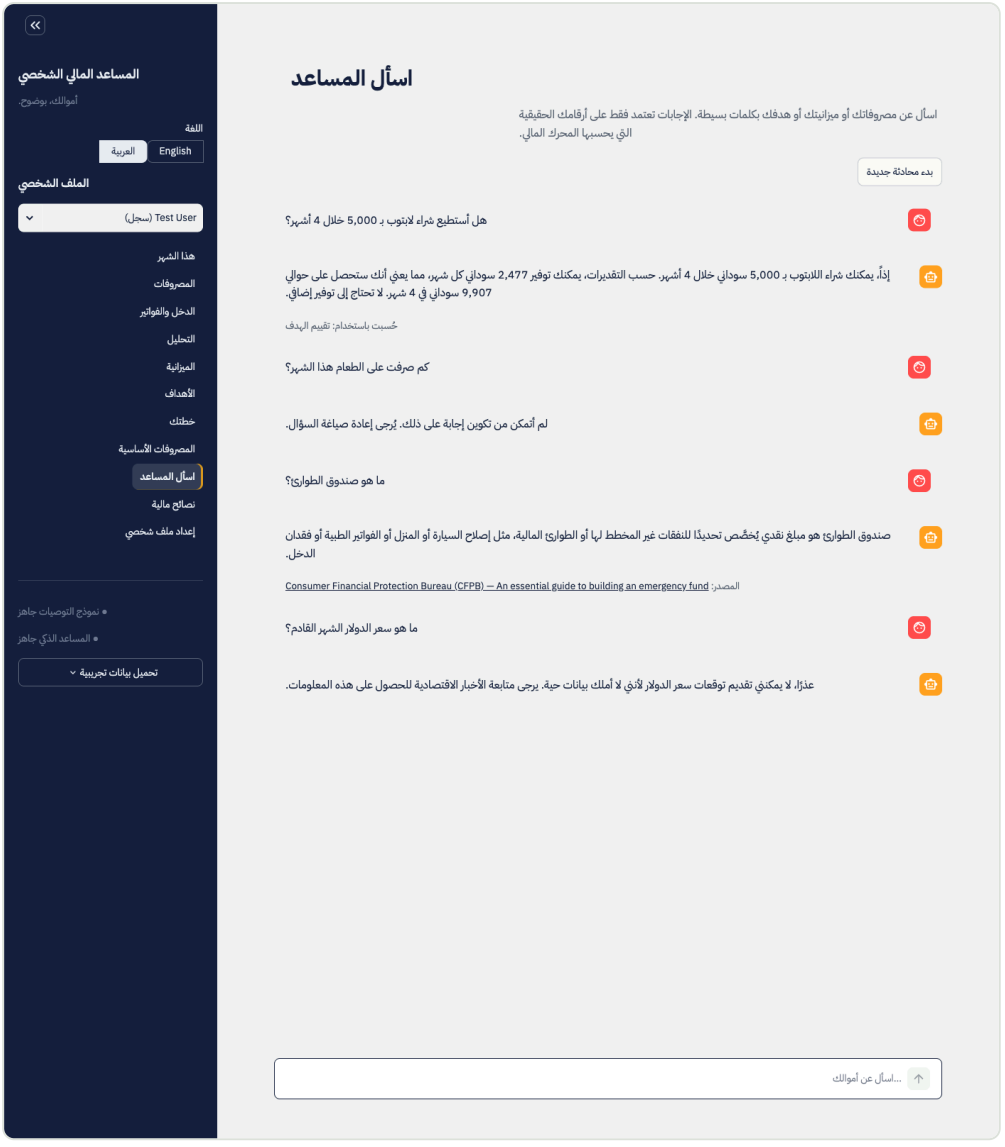
## UAT-28 The assistant in Arabic

1. Switch to العربية and ask, in one conversation: can I afford a 5,000 laptop in 4 months; how much did I spend on food this month; what is an emergency fund; what will the dollar rate be next month.

**Expected:** four correct answers in Arabic. **Actual:** three of four are correct:

- the laptop answer gives 2,477 a month, about 9,907 by then, and "no extra saving needed";
- the emergency fund answer quotes the verified Arabic note and names the CFPB;
- the dollar-rate question is refused.

The food question gets "لم أتمكن من تكوين إجابة... يُرجى إعادة صياغة السؤال" (could not answer, please rephrase). Fail. The cause is the model: Qwen 2.5 3B chooses the wrong tool for this Arabic question (total spending instead of food). The app now says so clearly instead of showing an empty answer or a wrong number. Two smaller points: amounts are written as "سوداني" rather than "ج.س", and before defect D-09 was fixed every Arabic question got an empty reply. A larger model (Qwen 2.5 7B, about 4.7 GB) is the expected fix.




Assistant answers in Arabic

## 4. Defects

Sixteen defects were found during UAT, all fixed and re-tested. Seven were in the interface (D-01 to D-07). Nine appeared only when the real AI model was installed (D-08 to D-16): the earlier tests used a scripted stand-in for the model, which always behaves. The most serious were D-10, a general question answered from memory with an invented source, and D-11, no tool able to answer "this month" questions. One problem stays open as a limit of the 3B model (UAT-28).

| ID | Severity | Found in | Defect | Fix | Status |
|---|---|---|---|---|---|
| D-01 | High | UAT-15 | Dragging a goal on the priority board did not save the new order: the board's single-list mode never reported the change | A one-column board in multi-column mode, the same as the working essentials board; verified by dragging | Fixed and re-tested |
| D-02 | Medium | UAT-04 | After **Add expense**, the form kept its values, inviting duplicate entries | The save runs in a callback that empties the form | Fixed and re-tested |
| D-03 | Low | UAT-08, 09, 12 | The income, bill, goal and add-savings forms also kept their values | Each form's fields reset after a successful save | Fixed and re-tested |
| D-04 | Low | UAT-17 | Chart labels were cut off ("Reduce entertai...") | Wider label space on bar charts | Fixed and re-tested |
| D-05 | Medium | UAT-22 | In Arabic, an English goal name with its amount displayed scrambled on the priority board | Unicode direction isolates around the name and the amount | Fixed and re-tested |
| D-06 | High | UAT-21 | After switching language and changing page, the app jumped back to user 1 | The chosen page and profile are kept in the session and pushed into the widgets | Fixed and re-tested |
| D-07 | Low | Log review | Streamlit warned that `use_container_width` will be removed, so a future version would break the app | Replaced by `width="stretch"`; Streamlit 1.50 or newer required | Fixed and re-tested |
| D-08 | High | UAT-25 | The assistant wrote amounts in dollars ("$2,476.67") | SDG named in every tool result; stricter instruction; any dollar amount rewritten as SDG / ج.س | Fixed and re-tested |
| D-09 | High | UAT-28 | Every Arabic question got an empty reply (the 3B model stops at once when Arabic text meets the tool list) | Arabic questions, including earlier ones in the conversation, are framed with a short English instruction | Fixed and re-tested |
| D-10 | High | UAT-25 | A general question was answered from the model's memory, citing an invented website | The app searches the knowledge base before the model answers; links that are not verified sources are removed | Fixed and re-tested |

| ID | Severity | Found in | Defect | Fix | Status |
|---|---|---|---|---|---|
| D-11 | High | Benchmark | No tool could answer "how much did I spend on food this month": the spending tools only knew all-time totals | Spending tools default to this month and also give each month, subcategories and month-to-month changes | Fixed and re-tested |
| D-12 | Medium | Benchmark | Knowledge notes added to questions about the user's own numbers distracted the model; an extra argument failed the whole tool call | Notes only for general questions or strong matches; unknown arguments ignored | Fixed and re-tested |
| D-13 | Medium | UAT-25 | The affordability answer contradicted itself ("save an additional 2,476.67 a month" while on track) | The goal assessment returns a plain one-sentence verdict for the model to use | Fixed and re-tested |
| D-14 | Medium | UAT-28 | An empty answer bubble; a failed answer was then copied in later replies | Ask the model once more, then show a clear "please rephrase" message; failed exchanges left out of the history | Fixed and re-tested |
| D-15 | Low | Benchmark | Answers changed from run to run; "restaurants" (lower case, plural) found nothing | Temperature 0 with a fixed seed; loose category matching | Fixed and re-tested |
| D-16 | Low | UAT-25 | "Start a new conversation" appeared between chat messages | Moved above the conversation | Fixed and re-tested |

**Open items (not defects)**

- UAT-28 fails on one Arabic question: Qwen 2.5 3B picks the wrong tool for "how much did I spend on food this month" in Arabic. Expected fix: Qwen 2.5 7B (about 4.7 GB).
- Dragging on a phone was not tested.
- The demo amounts (for example 6,000 SDG income) were not rescaled to current Sudanese prices. They illustrate the method only.

## 5. Summary and sign-off

The system is ready for acceptance, with one known limitation. 27 of 28 cases pass. The failing case, UAT-28, is one Arabic chat question that the small 3B model cannot answer; the app says so clearly rather than giving a wrong number. All 16 defects found are fixed and re-tested.

| Result | Count |
|---|---|
| Passed | 27 |
| Failed (known limit of the 3B model) | 1 |
| Blocked | 0 |
| Defects found and fixed | 16 |
| Automated tests passing | 271 |

| Role | Name | Decision (accept / accept with conditions / reject) | Signature | Date |
|---|---|---|---|---|
| Tester | | | | |
| Project team | | | | |
| Supervisor | | | | |